Target Echo Detection Based on the Signal Conditional Random Field Model for Full-Waveform Airborne Laser Bathymetry

Kai Guo, Qingquan Li, Chisheng Wang, Qingzhou Mao, Yanxiong Liu, Yongzhong Ouyang, Yikai Feng, Jiasong Zhu, and Anlei Wu

***Abstract*— Airborne laser bathymetry (ALB) systems with digital full-waveform signal collection can obtain the corresponding temporal positions from several backscattering surfaces by laser beam irradiation. This information can help describe the multielevation structures of the target and explore the echo signal attenuation response in different nonuniform mediums during laser propagation. Therefore, a full-waveform echo signal is quite practical for integrated water–land detection. However, the wavelength used in the ALB system is generally in the visible band range of 470–580 nm, and the received signal is constantly interfered with by many nontarget factors, such as imperfections in the receiving channel or the strong scattering from the transmission medium. The conventional processing method transforms nontarget interference into noise point cloud filtering or classification extraction, enabling the detection of a single surface or regular geometry. The accuracy of the identification and extraction for multielevation target surfaces echo signal is always reduced due to the significant noise signal intensity. We proposed a signal component detection method by constructing the echo signal feature functions and the conditional random field (CRF) model based on the full-waveform decomposition. The processing result for actual measurement data verified that the CRF strategy can effectively reduce the uncertainty of target surface detection. Compared with the single-beam echo sounder, the root-mean-squared errors of the elevation deviation underwater were reduced by 3.2 and 4.9 cm in the two different experimental areas.**

***Index Terms*— Airborne laser bathymetry (ALB), conditional random field (CRF), full-waveform signal, land and water integration point cloud.**

I. Introduction

THE development of integrated land and sea exploration technology is an important task for geographical environment monitoring in coastal areas and is also an effective model system for improving the flexibility and efficiency of remote sensing equipment [1], [2]. Compared with the limitations of shipborne measurement methods and traditional optical remote sensing technology in unnavigable areas, airborne laser bathymetry (ALB) avoids the adverse effect of a complex environment on the effectiveness of target monitoring [3], [4]. This performance primarily relies on the transmission properties of blue–green lasers (470–580 nm) in a water area and the active detection mode adopted by an ALB system [5]. Moreover, an ALB system has advantages in describing the features of underwater and shallow sea area topography, constructing 3-D underwater topographic models, and extracting the optical features of water bodies [6].

The ALB system using full-waveform echo signal technology can record the response characteristics of the backscattering cross section and the information of the laser propagation medium quality [7], [8], providing data that serve as a reliable basis for the monitoring and analysis of target features and changes [2], [9]. The detection capability can be enhanced by the optimization of the receiver strategy in the complex regional environment, but this approach is usually insufficient for enhancing the extraction effect of the multielevation structure [10]. Therefore, echo signal processing and point cloud generation methods will be beneficial for extracting the land, underwater topography, and environment changes [2].

In actual signal processing, the prior echo properties of specific signal segments in the full-waveform echo signal before generating point cloud results with structured features, such as ground, water surface, water bottom, and various ground objects, are usually unknown. Therefore, the processing for the

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detection of the signal intensity change in the full-waveform signal and its use as the target echo temporal position is instrumental in the rapid generation of the laser point cloud for an online system [11]. However, real-time changes in the natural environment, the nonuniformity of the laser propagation medium, and errors by the signal receiving system usually give rise to intensity attenuation, noise interference, and overlapping signal components in the waveform signal [12], [13], [14]. The interference makes the target echo signal component difficult to distinguish within the full waveform, diminishing the target extraction accuracy from a multielevation echo surface in near-shore and shallow water areas [13], [15].

Relevant studies have shown that the echo signal can be regarded as the convolution of the transmitted waveform and the target echo signal characteristics [16], [17], [18]. Since the echo energy of the transmitted pulse can usually be approximated to the Gaussian distribution which the model is concise [19] and does not require excessive prior information for the signal waveform segment in the processing process, the waveform decomposition method based on the Gaussian mixture model is widely used in the ALB echo signal processing and analysis [20], [21], [22]. Based on waveform decomposition waveform decomposition, it is of great practical significance to accurately determine the multielevation surface echo components in the 1-D full-waveform echo signal to improve the actual detection capability of the system. Signal denoising and target echo component identification can reduce the adverse effects of nontarget signal components in the integrated water–land point cloud generation.

Saylam et al. [23] studied the use of moving smoothing to filter echo signal noise. Their adopted method has high processing efficiency and a good effect on random noise processing. Wagner and Melzer [20] proposed signal filtering based on the average square difference function (ASDF). This method calculates the correlation values by matching the transmitted waveform and the echo waveform. Then, the noise components with low correlation in the signal can be suppressed. Liu et al. [24] proposed a depth-adaptive matched filter based on Monte Carlo simulations that can solve the deformation problem of underwater signals at different depths to a certain extent. This denoising method is more effective than the moving smoothing method, but the processing requires the laser emission signal as *a priori* information. Moreover, the decomposition of echo signals by wavelet multiscale transform and suppression of the nontarget components is also a widely studied field. Qi et al. [25] realized wavelet self-adaptive denoising by adjusting the scale factor of the derivable threshold function based on the waveform signal-to-noise ratio. Song et al. [26] combined the wavelet threshold and an empirical model to filter the component coefficients and achieved a better effect than conventional wavelet threshold denoising. Zhao et al. [27] proposed an adaptive threshold selection (ATS) wavelet threshold for underwater signal detection to improve the performance of the random noise reduction by adaptively determining the threshold for each full waveform. In summary, the difficulty of wavelet analysis for nontarget noise suppression lies in the determination of the wavelet threshold in the complex detection environment.

For the determination of the target echo component, the target echo classification method based on the waveform characteristics of the green laser channel is significant as a reference method. Collin et al. [28] measured the target echo component features by calculating 21 statistical parameters. Wilson et al. [29] analyzed the relationship between waveform features and coral reef morphology. Based on the waveform decomposition, Su et al. [30] proposed an identification method for coral reef classification and achieved a good effect on signal component identification. Waveform component extraction can obtain good processing results by analyzing the echo waveform features as identification conditions, but it usually requires a large amount of data as statistical reference data or faces the problem of the insufficient coupling relationship of the waveform components. The difficulty of extracting complex waveform features also restricts the use of this method.

The single-wavelength ALB system is sensitive to strong scattering in the propagation medium, and high-density nontarget points interfere with the effect of data processing based on the spatial position. Some studies transformed the target surface detection into spatial filtering or classification of point clouds to avoid the uncertainty of the 1-D echo signal of the visible laser [31]. However, spatial filtering algorithms generally rely on several fundamental principles, such as morphology, surface fitting, and irregular triangulation [32]. Therefore, such a method is effective for the monolayer target surface, and the extraction performance for multielevation structures or overlapping targets with complex terrain exhibits significant uncertainty in the processing results [33], [34].

In addition, when the influence of nontarget signals is significant, frequent manual intervention would be even necessary. Therefore, many artificial intelligence methods have been introduced for point cloud processing, such as point cloud clustering [35], conditional random fields (CRFs) [36], and artificial neural networks [37]. Comprehensive identification algorithms are becoming important for improving system performance and reducing the manual workload. These methods generally rely on the position and spatial structure of a point cloud and are effective in application to inland target detection based on infrared light detection and ranging (LiDAR) [37], [38]. However, these algorithms face additional challenges when the detection laser used by the LiDAR system is in the visible spectrum [39], [40].

In this article, we proposed an intelligent approach to extracting the features of echo components and then verified the accuracy of target signal recognition in ALB system processing results. The content of this article includes the following.

1) The interference of nontarget signals and echo component superposition for target echo extraction are studied, and the waveform processing strategy and the point cloud generation model have been designed and implemented.

2) Based on waveform decomposition, the echo feature functions of unary and pairwise potentials are constructed for CRF modeling. Then, we design a model training strategy that suits the integrated signal processing for the overall regional land and seafloor.

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3) By using a self-developed ALB system and the measured data, we verify the effectiveness of the target echo component identification based on the CRF model and analyze the characteristics of the method with an actual application of the system.

II. Methodology

The distance estimation between the receiving system and target is a focus of airborne laser echo signal processing. A general system primarily uses the time-of-flight (TOF) mode to calculate the distance L between the target and the transmitter by recording the time interval between the laser emission and the target signal return

$$L = \frac{1}{2}ct \tag{1}$$

where c is the speed of the laser in the propagation medium. For an ALB system, the value of t and its accuracy are directly related to the determination of the target position. Therefore, the final purpose of feature extraction for a target echo is to determine the propagation distance of the echo signal. According to the analysis, the echo signal $p(t)$ can be decomposed into the following function:

$$p(t) = \sum_{\tau=1}^{m} f_\tau(t) + n(t), \quad n(t) \sim N\left(0, \sigma^2\right) \tag{2}$$

where $n(t)$ is the noise component, $f_\tau(t)$ is the subfunction, and m is the number of $f_\tau(t)$. In this article, we use the Gaussian half-wavelength progressive decomposition (GHPD) method to decompose the echo signal and the basic echo signal parameters [9]. The following is the full-waveform ALB data processing procedure (Fig. 1) based on the CRF of the echo signal features.

Before extracting the target echo signal, we directly preprocess the full-waveform signal obtained by the ALB system, including the selection of the target echo segment and low-pass filtering. This approach helps to place the processing focus on the target echo signal segment and to suppress the noise $n(t)$. We use a Gaussian mixture model to decompose the signal waveform and extract the feature of each echo component [9]. On this basis, a CRF model will be established to eliminate the interference of nontarget echo signals and identify the target echo component in the whole detection area. Then, a point cloud can be calculated with the POS data and the corresponding laser emission direction according to the obtained target echo temporal position. The data processing scheme aims to reduce the manual work needed for nontarget point cloud removal and improve the multiple target detection performance of the ALB system.

A. Signal Preprocessing and Echo Features

1) Signal Preprocessing: Many ALB systems are designed with receiving channels for separately recording the transmit pulse signal and the echo signal of the laser. Calibration of the laser emission time corresponding to the current received signal is the basic purpose of the transmit signal. A full-waveform signal can be expressed as a 1-D intensity vector in the unit of signal sampling interval using analog-to-digital conversion.

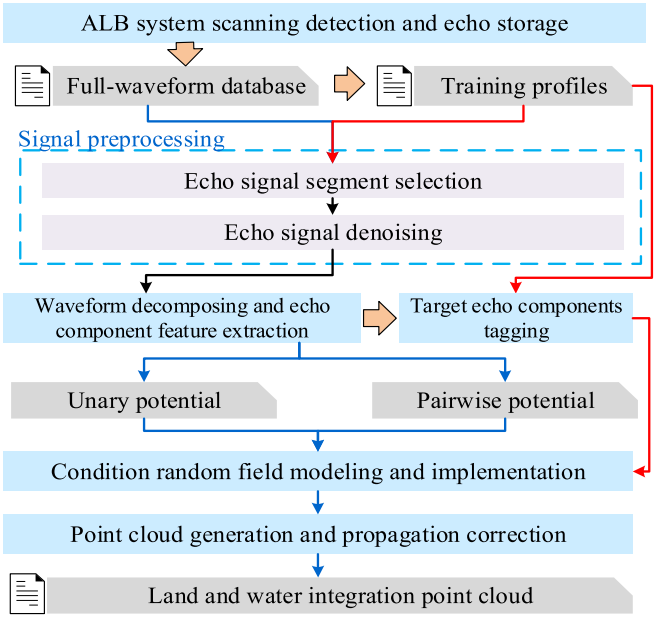

Fig. 1. Data processing procedure.

To ensure that the space detection range is expanded as much as possible under the condition of the system detection capability, some ALB systems usually set a longer time sequence to record the intensity data of the echo signal. For example, the waveform data length of the Aquarius system is 288 ns, and the iGreena system used in the experiment in this article records the full-waveform sequence of up to 720 ns (1.25 GHz). Signal preprocessing includes the following two aspects.

a) Echo signal segment selection: Extracting the signal segment that reflects the target echo according to the waveform signal characteristics helps reduce the data redundancy in the original signal collected by the receiving subsystem. This ensures that the implementation of the signal processing method focuses on the signal segment containing the target echo. For the problem of selecting echo signal segments, we adopt a processing method based on the local no stationary characteristics of the target echo signal. This method can not only determine the echo signal segment accuracy but also maintain information integrity.

b) Low filter and signal denoising: The divergence angle and emission intensity of laser beams used by ALB systems are usually different from their respective design goals. In addition, the full-waveform signal is generally affected by the complex environment and circuit of the system itself and contains considerable signal noise. To improve the accuracy of waveform processing, it is necessary to address the adverse effects caused by signal noise and suppress the obvious nontarget echoes in the full-waveform data before waveform processing. The low-pass filter will be used to minimize the interference of random noise in the signal to subsequent waveform data processing.

We use a mixture of Gaussian functions to decompose the full-waveform signal. The time-domain echo signal from the seafloor represented by this Gaussian function is

$$y_G(n) = A \exp\left(-\frac{(n - n_{\max})^2}{2b^2}\right) \tag{3}$$

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where A is the amplitude, T is the sampling period, $n_{\max} = (t_{\max}/T)$ indicates the sampling rate at the peak of the wave, σ is the width of the echo signal, and $b = (\sigma/T)$ can be used as a control parameter. After the Fourier transform, formula (3) can be expressed as follows:

$$Y_G(v) = b\sqrt{2\pi}\exp\left(-\frac{b^2T^2v^2}{2}\right). \tag{4}$$

The b value is related to the physical and chemical properties of the seafloor. Some studies have shown that more than 90% of sea bottoms have the characteristics of $b > 4.8$. According to the empirical relationship, $v_c = 0.683/b$ and $v_c > 0.14$ can be obtained. On this basis, an finite impulse response (FIR) digital window filter will be used for denoising the full-waveform signal [41]

$$F_{\text{hw}}(n) = -\frac{1}{\pi}\left(\frac{\sin n\pi v_c}{n^2} - \frac{\pi v_c \cos n\pi v_c}{n}\right) \tag{5}$$

where $n = 1, 2, \ldots, K$, in which K is the number of signal intensities contained in the filter window.

2) Full-Waveform Decomposition: The full-waveform echo signal is considered the superposition of the echo signal from the different reflection cross sections and the random noise in the laser propagation path. We assume that the energy of the returned echo pulse obeys a Gaussian distribution; then, according to formula (2), the full-waveform data can be expressed as follows:

$$\begin{aligned} W(t) &= \sum_{i=1}^{m} x_i(t, p_i) + n(t) \\ &= \sum_{i=1}^{m} A_i e^{-(t-\mu_i)^2/2\sigma_i^2} + n(t) \\ p_i &\sim (A_i, \mu_i, \sigma_i), \quad n(t) \sim N\left(0, \sigma^2\right) \end{aligned} \tag{6}$$

where $x_i(t)$ is the time response function of each echo component and $n(t)$ represents the Gaussian noise. The three parameters (A_i, μ_i, σ_i) of each Gaussian function are related to the properties of the reflection cross section. The parameters represent the amplitude, reflection time, and signal width of the echo component. Therefore, the essence of Gaussian function decomposition is to determine the parameter value of each echo component. Signal decomposition can decompose laser echo components into multiple factors and has great significance for target detection and propagation process analysis. After preprocessing, we used a mixed progressive Gaussian function to decompose the waveform. Signal decomposition and target detection are implemented without much prior knowledge of the target position. The time position of the target echo component in full-waveform data is usually affected by the superposition of different components. Therefore, we adopt the GHPD method to decompose the full-waveform signal. The GHPD method separates the echo components in sequence according to the receiving order of the echo signal, which can reduce the peak value offset caused by the superposition among the echo components.

To improve the fitting accuracy of echo components, the Levenberg–Marquardt (LM) method is used to optimize the parameters of the Gaussian components and improve the decomposition accuracy of each echo component. As the LM method is based on the shape of the full-waveform signal features, it can achieve better fitting accuracy. The iterative formula of the LM algorithm is given as follows:

$$\begin{aligned} \hat{p}_i = p_i^{(0)} + \left[J^T\left(t, p_i^{(0)}\right)J\left(t, p_i^{(0)}\right) + \lambda E\right]^{-1} \\ \cdot J^T\left(t, p_i^{(0)}\right)\left[y - x_i\left(t, p_i^{(0)}\right)\right] \end{aligned} \tag{7}$$

$$\left\|\hat{p}_i - p_i^{(0)}\right\| < \varepsilon \tag{8}$$

where $J(t, p_i)$ is the Jacobian matrix of the Gaussian component, λ is a damping coefficient, and $\|\hat{p}_i - p_i^{(0)}\|$ is used as the criterion to judge the termination of the iteration. If the value $\|\hat{p}_i - p_i^{(0)}\|$ is larger than the threshold ε, $\hat{p}_i$ will be calculated as the new $p_i^{(0)}$ until the termination condition (8) is established.

3) Target Echo Features: After decomposing the full-waveform signal based on (6), each echo component will show some basic features, the most prominent of which are the unary features, i.e., amplitude, pulsewidth, and time position. The amplitude reflects the intensity of the echo signal, which is related to the reflectivity of the target surface and is significantly affected by propagation attenuation effects such as scattering and absorption by the medium. The time position of the echo component corresponds to the propagation distance of the echo signal. Target echoes that are closer along the propagation path will often overlap with each other, which significantly affects the performance in distinguishing different targets. We will analyze this phenomenon in the following. In addition, the width of the echo component usually shows a broadening compared with the emitted pulse, and the broadening degree reflects the attenuation effect when it propagates in the medium.

Taking the iGreena ALB system, developed by Shenzhen University, as an example, the receiving system can simultaneously record the waveform of an emitted pulse and the echo full-waveform signal in a single laser shot. Fig. 2 shows the result of the emitted waveform fitting and the corresponding echo waveform decomposition.

Fig. 2(a) shows the Gaussian fitting of the emitted waveform. According to the Gaussian model represented in formula (6), the parameters of the amplitudes, wave widths, and laser emission times can be obtained. The echo width will be expressed as $2\sigma_i$ in this article. Fig. 2(b) shows the result of echo waveform decomposition.

The echo signal received by the system is the convolution between the emitted pulse and several response functions, such as medium attenuation or target cross section backscattering. Therefore, the echo component after decomposition of the full-waveform signal correlates to the emitted waveform. The emitted waveform can be used in the evaluation of the target echo component. Furthermore, the mutual superposition of the echo components causes the echo components to be unrecognizable. The positional relationship between the echo components must be considered in the target recognition process when the time positions of the adjacent echo components are relatively close. In the same scanning operation, there is a coupling relationship between the different echo component features. Due to the

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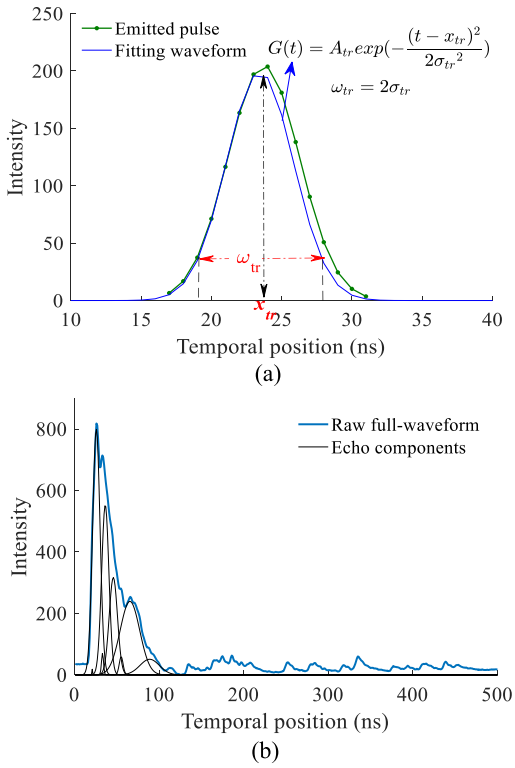

Fig. 2. (a) Emitted waveform and (b) echo full-waveform signal of the iGreena system.

complexity of the environment of the detected target area, the point cloud generated by the single condition selection method will also be different in the target area covered by different routes. For complex coastal topographic scanning, some methods even require extensive manual processing in actual applications, which will cause the data processing efficiency of the ALB system to decrease.

In this article, we propose to use echo features to realize intelligent identification of the target echo signal, particularly for the resolution of the target echo in the case of overlapping waveforms, and improve the detection effectiveness of the system.

B. CRF Modeling

The next goal in waveform processing is to identify the target echoes from the decomposed result. Based on waveform decomposition, the parameters of the echo component can compose a Markov chain in the order of signal receipt. The hidden Markov model (HMM) and the maximum entropy Markov model (MEMM) need to be limited by the independence assumption, which is difficult to satisfy for actual collected data. The CRF model is an undirected graphical model with an underlying graph. The echo signal features can be normalized for the global optimal value calculation in CRF modeling to reduce the labeling bias caused by a single node [42].

In addition, the coupling between components is considered in the feature function construction. Therefore, in this section, we transform the identification of a target echo component to a conditional probability of the given observation sequence problem, as shown in Fig. 3.

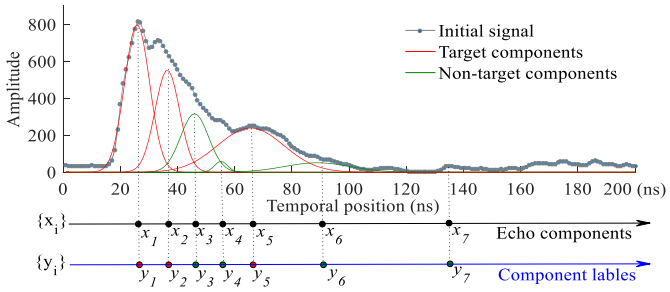

Fig. 3. Condition random field of the echo component feature.

In signal processing, each node x_i corresponds to a waveform component. We assign y_i to all nodes based on the set $\{x_i\}$. $\{y_i\}$ is the set that contains the attitude tag of a target or nontarget. $\{y_i\}$ has the same number of elements as $\{x_i\}$. Moreover, the graph edges represent the relations between the adjacent nodes. The key to target echo identification based on the CRF algorithm lies in the construction of a feature function and the estimation of model parameters.

1) Unary Potential: Waveform decomposition produces many echoes along the laser propagation path, but only parts are related to targets, while the others may be caused by noise. In this section, we will construct the feature functions based on the echo width, the echo component amplitude, the time position of adjacent components, and the similarity to the emitted signals. On this basis, we will establish the CRF model for tagging a target echo component.

a) Echo component width: Since the pulse will be stretched by the scattering during laser propagation, the target echo width should be larger than the width of the emitted pulse. We filter the target echo component by the waveform width of the emission pulse

$$w_\tau > w_{\text{tr}} \tag{9}$$

$$S_1(x_\tau) = \begin{cases} 1, & W(x_\tau) > w_{\text{tr}} \\ 0, & \text{others} \end{cases} \tag{10}$$

where $W(x_\tau)$ is the width of x_τ and w_{tr} is the width of the emitted pulse. If the echo component width is smaller than the emitted pulse, it can be classified as a nontarget response, and $S_1(x_\tau) = 0$.

b) Echo component intensity: During the echo propagation process, the attenuation of the echo energy is usually exponential [43], [44], [45]. To facilitate the use of signal strength to identify the components, we remove the influence of the exponential decay trend in the signal on the intensity change analysis (Fig. 4). Therefore, after extracting the first echo component, we use an exponential function to fit the remaining segments and take the fit exponential function as the approximate trend of the echo signal attenuation [17]. If the signal amplitude of the echo component is outside the noise confidence interval, it may be due to the target response after removing the above exponential trend. An effective target return is supposed to be significantly larger than the background noise. We estimate the noise standard deviation σ_n using the data at the tail of the waveform. If the signal magnitude is three times larger than σ_n, we can extract it from noise with 99.73% confidence probability.

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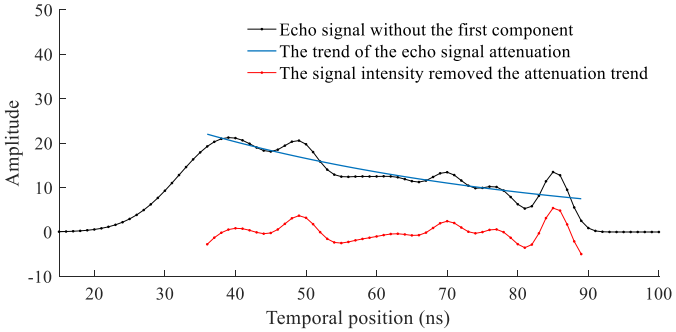

Fig. 4. Removal of the exponential decay trend after the extraction of the first echo component.

Let $\{x_\tau|\tau = 1, 2, \ldots, N_x\}$ be the echo component and $\tilde{A}(x_\tau)$ represent the intensity value of each component that remained from the attenuation trend removal. The feature function for echo signal amplitude is

$$S_2(x_\tau) = \begin{cases} 1, & \tilde{A}(x_\tau) \in [-3\sigma_n, \ 3\sigma_n] \\ 0, & \text{others.} \end{cases} \tag{11}$$

c) Echo signal similarity: According to our design, an emitted pulse waveform is recorded synchronously in the seed light channel. The received target response should have a waveform shape similar to that of the emitted pulse. We used the ASDF to calculate the correlation between the received waveform and the emitted pulse. Echo locations with a low ASDF, therefore, have a low probability of being a target response [20].

The ASDF is a time delay estimation based on the correlation between the reference signal and the received signal. Therefore, the ASDF method can suppress the negative effect of noise in an echo waveform. In this section, we attempt to estimate the similarity between the received waveform and the emitted waveform and use it as the basis for judging when reflection occurs.

Let x_{tr} and x_τ be the transmitting waveform and the receiving waveform, respectively. The correlation coefficient R_{ASDF} of the ASDF can be calculated as follows:

$$R_{\text{ASDF}}(\tau) = \frac{1}{N}\sum_{k=1}^{N}[x_{\text{tr}}(kt_T) - x_\tau(kt_T + \kappa)]^2 \tag{12}$$

where t_T is the sampling interval during signal receiving and $\kappa = -Nt_T, (-N+1)t_T, \ldots, Nt_T$ represents an integer multiple for t_T.

The temporal position corresponding to the extreme value in the correlation coefficient sequence is the position where a target echo may appear. Therefore, to avoid the effect of noise in the signal and improve the accuracy of the target judgment, the temporal positions corresponding to the first three largest extremums of the correlation coefficient are extracted as the possible target echo components

$$S_3(x_\tau) = \begin{cases} 1, & \min\left(|\hat{R}_{\text{ASDF}}(t) - T(x_\tau)|\right) < w_{\text{tr}} \\ 0, & \text{others} \end{cases} \tag{13}$$

where $\hat{R}_{\text{ASDF}}(t)$ is the maximum value in the correlation coefficient sequence obtained by the ASDF method. Although the correlation coefficient sequence can eliminate the influence of noise, it also causes a certain delay in the corresponding position of the echo. We still use w_{tr} as the confidence radius.

2) Pairwise Potential: The unary features play a significant role in distinguishing the target echo from the adjacent components. Conventional condition determination compares the intensity, width of the echo components, and relative time distance between the two as separate signal features. However, the different signal features of adjacent echo components have coupling effects in target echo identification. We have studied target echo extraction from adjacent echo components under the condition of waveform superposition. The pairwise features between the adjacent components will be introduced, which will be helpful for improving the accuracy of target echo signal identification.

First, we constructed indicators to quantify the relationship between the features of adjacent waveform components.

a) Amplitude multiple: The ratio of the amplitudes of adjacent echo components

$$m_{a_\tau} = \frac{\max\left[A(x_{\tau-1}), A(x_\tau)\right]}{\min\left[A(x_{\tau-1}), A(x_\tau)\right]}. \tag{14}$$

b) Width multiple: The ratio of the waveform widths in units of the half-width of the transmitted pulse

$$m_{w_\tau} = \frac{2\max\left[W(x_{\tau-1}), W(x_\tau)\right]}{w_{\text{tr}} \cdot \min\left[W(x_{\tau-1}), W(x_\tau)\right]}. \tag{15}$$

c) Temporal distance: The time distance between the peak positions of adjacent echo components based on the half-width of the transmitted pulse

$$m_{t_\tau} = \frac{2}{w_{\text{tr}}}|T(x_\tau) - T(x_{\tau-1})|. \tag{16}$$

We perform a simulation experiment to study the distinguishability between adjacent Gaussian components and sequentially analyze the effect of the wave width and relative time distance of adjacent echo components under different intensity multiples. Fig. 5 shows the case in which the simulated adjacent Gaussian components are superimposed. Waveform peak detection based on the first-order difference is used to test whether the two components can be extracted. Fig. 5 shows that the two simulated components cannot be effectively distinguished, while the components can be identified from the superimposed waveform.

The distinguishability of adjacent echo components will change with the three indicators in $[m_{a_\tau}, m_{w_\tau}, m_{t_\tau}]$. In Fig. 6, we detect the intensity mutation of the signal superposition waveform with different values of m_{a_τ} and plot the positions as the critical points corresponding to the first occurrence of more than two mutations in the rectangular coordinate space composed of the temporal distance and width multiple values. The curve fitting of the points also shows the trend of the changes in the distinguishability. Fig. 6(a) shows that the curves have some similar trend characteristics $[m_{t_\tau}, m_{w_\tau}]$. As shown in Fig. 6(b), taking $m_{a_\tau} = 3$ as an example, each curve can be divided into three segments: $(0, m_{\text{ts}})$, $[m_{\text{ts}}, m_{\text{te}}]$, and (m_{te}, ∞), and the curve is divided into five regions.

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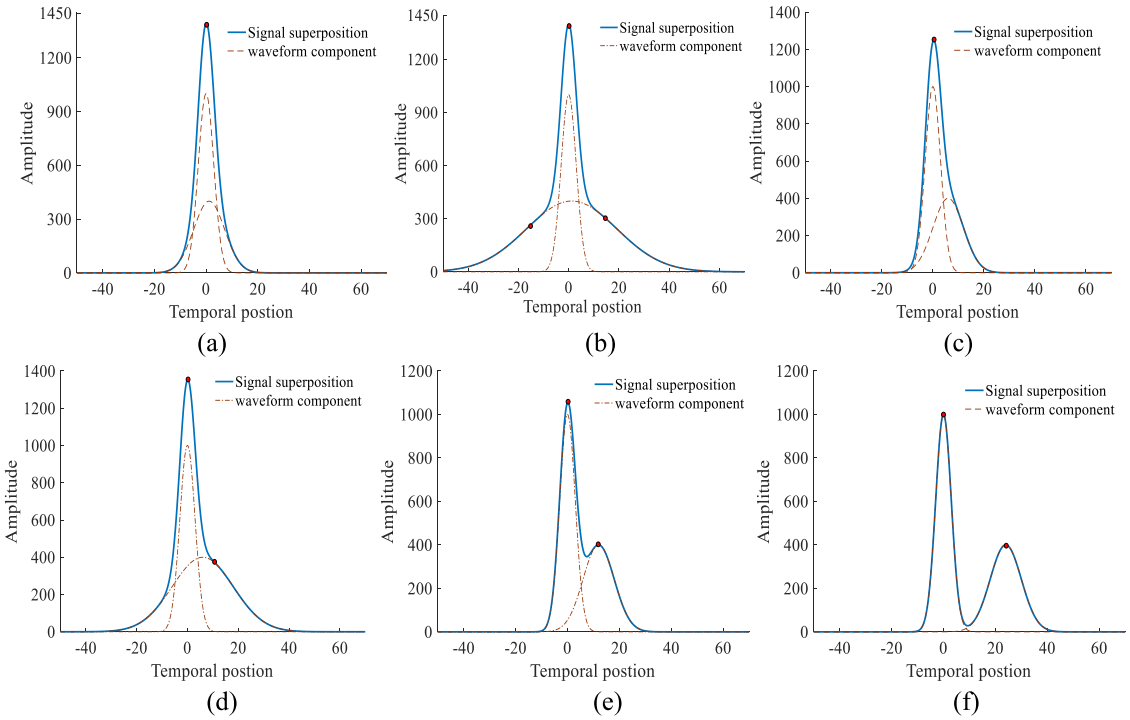

Fig. 5. Component distinguishability with echo waveform component superposition (the red point reflects the temporal position of the amplitude mutation that can be found by using the first-order difference method, $m_{a_\tau} = 1000/400$). (a) $m_{w_\tau} = 2$ and $m_{t_\tau} = 0.2$. (b) $m_{w_\tau} = 4$ and $m_{t_\tau} = 0.2$. (c) $m_{w_\tau} = 2$ and $m_{t_\tau} = 0.8$. (d) $m_{w_\tau} = 4$ and $m_{t_i} = 0.8$. (e) $m_{w_\tau} = 2$ and $m_{t_\tau} = 1$. (f) $m_{w_\tau} = 2$ and $m_{t_\tau} = 4$.

$T(x_\tau) \in (0, m_{\text{ts}}]$: If (m_{t_τ}, m_{w_τ}) is located in area ①, the number of components obtained by signal detection will be greater than 2. As shown in Fig. 5(b), since the time interval between adjacent components is small and the width difference is large, the time position of the detected signal strength mutation is different from the real temporal position. Thus, adjacent individual components cannot be identified correctly. In Fig. 6(b), when (m_{t_τ}, m_{w_τ}) is located in area ②, the echo components overlap with each other, but only one mutation can be detected. Therefore, waveform decomposition in this interval cannot clearly distinguish the two adjacent waveform components.

$T(x_\tau) \in (m_{\text{ts}}, m_{\text{te}}]$: Echo adjacent component identification will be affected by changes in different amplitudes and echo widths. Here, we use the polynomial model to fit the critical condition in $(m_{\text{ts}}, m_{\text{te}}]$

$$F = p_1 m_{t_\tau}^3 + p_2 m_{t_\tau}^2 + p_3 m_{t_\tau} + p_4 - m_{w_\tau} \quad m_{t_\tau} \in (m_{\text{ts}}, m_{\text{te}}] \tag{17}$$

where p_1, p_2, p_3, and p_4 are the model parameters. In this section, we consider the operability of the conditions and the accuracy of the processing results. In Table I, the amplitude multiple is divided into several equal discrimination intervals. The corresponding parameters of the discrimination interval model are shown in the table. Based on the model provided in the table, the distinguishability of an adjacent echo component can be estimated in a preliminary evaluation.

As shown in Fig. 6(b), when $F < 0$, (m_{t_τ}, m_{w_τ}) will be located in field ③ divided by the critical curve, and the adjacent components will be easily distinguished. When the position (m_{t_τ}, m_{w_τ}) is located in field ④, the echo signal is superimposed on one echo waveform and cannot be identified.

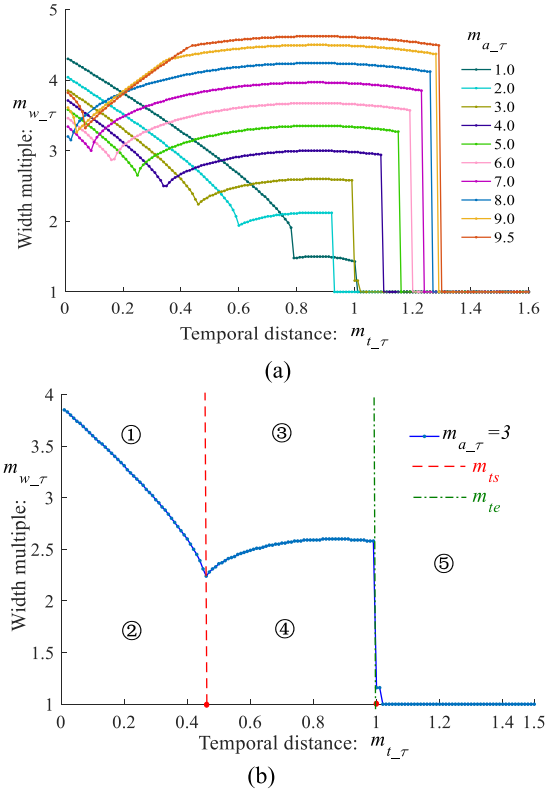

Fig. 6. Distinguishability between adjacent echo components and pairwise features. (a) Critical curves for different m_{a_τ} values. (b) Regions divided by the critical curve.

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TABLE I
CRITICAL CONDITION MODEL IN ($m_{\text{ts}}, m_{\text{te}}$]

Amplitude multiple	p_1	p_3	p_3	p_4	m_{ts}	m_{te}
$(-\infty,1.0]$	-0.001	-0.017	-0.012	1.500	0.79	1.00
$(1.0,1.5]$	0.003	-0.024	0.018	1.837	0.68	0.97
$(1.5,2.0]$	0.006	-0.025	0.038	2.102	0.60	0.92
$(2.0,2.5]$	0.010	-0.038	0.056	2.340	0.52	0.92
$(2.5,3.0]$	0.013	-0.052	0.059	2.575	0.46	0.99
$(3.0,3.5]$	0.016	-0.069	0.062	2.786	0.40	1.05
$(3.5,4.0]$	0.018	-0.079	0.067	2.981	0.35	1.09
$(4.0,4.5]$	0.023	-0.091	0.070	3.163	0.30	1.12
$(4.5,5.0]$	0.028	-0.106	0.073	3.335	0.25	1.15
$(5.0,5.5]$	0.028	-0.113	0.081	3.498	0.21	1.17
$(5.5,6.0]$	0.036	-0.128	0.086	3.653	0.16	1.19
$(6.0,6.5]$	0.037	-0.132	0.087	3.803	0.13	1.21
$(6.5,7.0]$	0.040	-0.143	0.091	3.947	0.09	1.23
$(7.0,7.5]$	0.014	-0.121	0.123	4.071	0.06	1.24
$(7.5,8.0]$	0.045	-0.152	0.091	4.221	0.02	1.24
$(8.0,8.5]$	0.073	-0.197	0.088	4.363	0.01	1.26
$(8.5,9.0]$	0.084	-0.223	0.089	4.500	0.04	1.28
$(9.0,9.5]$	0.085	-0.241	0.110	4.624	0.07	1.29
$(9.5,\infty)$	0.079	-0.257	0.147	4.738	0.10	1.30

$T(x_\tau) \in (m_{\text{te}}, \infty)$: In this interval, the influence of the coupling relationship between the echo component features on the echo component resolution is no longer significant, and the adjacent components can be distinguished.

We propose the following transition function, which can reflect the coupling effect of the adjacent echo component features:

$$T(x_\tau) = \begin{cases} 0, & F \geq 0 \ \cup \ m_{t_\tau} \in (0, m_{\text{ts}}) \ \cup \ T(x_{i-1}) = 0 \\ 1, & F < 0 \ \cup \ m_{t_\tau} \in (m_{\text{te}}, \infty) \ \cup \ T(x_{i-1}) = 1. \end{cases} \tag{18}$$

According to the condition analysis of the echo signal features, most irrelevant signals will be ruled out, and the component combination that meets the actual target echo signal can be identified, to determine their temporal positions on the water surface, the underwater target, and the seafloor.

3) Model and Implementation: After the full-waveform data decomposition, we calculated the conditional probability of the CRF model and its weight coefficients through model training. Based on the CRF model, target echo components can be extracted from the whole signal waveform. The algorithm can reduce signal echoes affected by several elevation-level surfaces or complex spatial structures.

$X = \{x_\tau | \tau = 1, 2, \ldots, N\}$ is the set of echo components in the same waveform after decomposition. $Y = \{y_\tau^{(i)} | i = 1, 2 \ ; \tau = 1, 2, \ldots, N\}$ is the corresponding label set for X. Y is built based on the echo attribute. The digital value of Y is $\{0, 1\}$. $y_\tau^{(1)} = 0$ represents a nontarget component, and $y_\tau^{(2)} = 1$ indicates that the corresponding echo signal is from the target surface.

Based on the above signal echo component feature functions (10), (11), (13), and (18), we construct an identification model according to the form of the CRF. The function and weight coefficient vector are given as follows:

$$\theta = [\theta_1, \theta_2, \theta_3, \theta_4]^{\text{T}} \tag{19}$$

$$F(y, x_\tau) = [f_1(y, x), f_2(y, x), f_3(y, x), f_4(y, x)]^{\text{T}} \tag{20}$$

where

$$f_k(y, x) = \sum_{\tau=1}^{N} f_k(y_{\tau-1}, y_\tau, x) \tag{21}$$

$$f_k(y_{i-1}, y_i, x_\tau) = \begin{cases} S_k(y_i, x_\tau), & k = 1, 2, 3 \\ T_k(y_{i-1}, y_i, x_\tau), & k = 4 \end{cases} \tag{22}$$

$$\theta = \begin{cases} \mu_k, & k = 1, 2, 3 \\ \lambda_k, & k = 4. \end{cases} \tag{23}$$

In formula (22), $T(y_{i-1}, y_i, x_\tau)$ is the transition function and $S_k(y_i, x_\tau)$ represents the state functions. λ_k and μ_k are the parameters of the CRF model. According to the features of the echo signal, the CRF model can be built as follows:

$$P_\theta(y|x) = \frac{\exp[\theta \cdot F(y, x)]}{\sum_y \exp[\theta \cdot F(y, x)]}. \tag{24}$$

The CRF parameter estimation and optimization method was used in the learning and training of the identification model. Therefore, we extract the observation data for model training first in actual processing. The extracted training data should contain different types of targets in the water and land space, which will help to distinguish related targets as much as possible.

The ultimate goal of parameter estimation is to calculate the extreme values of the following functions:

$$\begin{aligned} L(\theta) &= \sum_{x,y} \tilde{P}(x, y) \log P_\theta(y|x) \\ &= \sum_{\tau=1}^{N} \sum_{k=1}^{4} \tilde{P}(y_\tau, x_\tau) \theta_k f_k(y_\tau, x_\tau) \\ &\quad - \sum_{i=1}^{2} \log \sum_{\tau=1}^{N} \sum_{k=1}^{4} \tilde{P}\left(y_\tau^{(i)}, x_\tau\right) \theta_k f_k\left(y_\tau^{(i)}, x_\tau\right) \end{aligned} \tag{25}$$

where $\tilde{P}(x, y)$ is the joint empirical probability and the marginal empirical probability value, which can be calculated using the training data. Using a Taylor series to expand formula (24), we obtain the following iterative relationship:

$$\theta_{n+1} = \theta_n - H^{-1} g(\theta_n) = \theta_n - G_n^{-1} g(\theta_n) \tag{26}$$

$$g(\theta) = \sum_{x,y} \tilde{P}(x) P(y|x_\tau) f(x, y) - E_{\tilde{P}}(f) \tag{27}$$

where H^{-1} is the inverse of the Hessian matrix. According to the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm and the Sherman–Morrison model, H can be approximated by G_k. G_k can be calculated by the following iterative model:

$$G_{n+1} = \left(I - \frac{\varphi_n \Delta_n^{\text{T}}}{\varphi_n^{\text{T}} \Delta_n}\right) G_n \left(I - \frac{\varphi_n \Delta_n^{\text{T}}}{\varphi_n^{\text{T}} \Delta_n}\right)^{\text{T}} + \frac{\varphi_n \Delta_n^{\text{T}}}{\varphi_n^{\text{T}} \Delta_n} \tag{28}$$

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where $\varphi_n = g(\theta_{n+1}) - g(\theta_n)$ and $\Delta_n = \theta_{n+1} - \theta_n$. When $|\varphi_n| < \varepsilon$, the iterative calculation is terminated, and the parameters of the identification model for the scanning area have been confirmed.

After the CRF model of the full-waveform signal in the target area is established, the problem of identifying the target echo component can be transformed into that of labeling the hidden attributes of different waveform components in the same waveform signal.

According to the Viterbi algorithm, the nonnormalized probability of the attribute tag corresponding to each waveform component is calculated recursively

$$\delta_1(i) = \theta \cdot F_1\left(y_1^{(i)},\ x_1\right), \quad i = 1, 2 \tag{29}$$

$$\delta_\tau(i) = \max_{i=1,2}\{\delta_{\tau-1}(i) + \theta \cdot F_\tau(y_{\tau-1}^{(i)}, y_\tau^{(i)}, x)\},\ i = 1, 2. \tag{30}$$

By calculating the nonstandardized maximum value of each echo component for the different attribute tags, the attribute tag sequence of the nonnormalized maximum value is recorded in turn

$$\psi_\tau(i) = \arg\max_{i=1,2}\left\{\delta_{\tau-1}(i) + \theta \cdot F_\tau\left(y_{\tau-1}^{(i)}, y_\tau^{(i)}, x\right)\right\}, \quad i = 1, 2. \tag{31}$$

The recursion is continued until $\tau = N$. The maximum value $\max_{i=1,2} \delta_N(i)$ of the unnormalized probability and the optimal sequence $\{y_\tau\}$ of the attribute tagging can be obtained, where

$$y_\tau = \psi_{\tau+1}(y_{\tau+1}), \quad \tau = N-1, N-2, \ldots, 1. \tag{32}$$

By using the random field model to tag the echo component attributes in each full waveform, it is possible to extract the temporal position μ_i of the target echo component. Then, the propagation distance of the blue–green laser can be calculated for point cloud generation.

C. Point Cloud Generation

1) Above the Water Surface: We calculate the distance and direction from the ALB system to the target echo surface based on the signal waveform processing result and the optical–mechanical structure model. After fusing with the position and attitude data obtained by the system POS, the preliminary calculation for the footprints' geodetic coordinate is similar to the solution of the conventional LiDAR system. The coordinate conversion should be implemented according to the specific settings of each sensor

$$\vec{L} = \begin{bmatrix} x_L \\ y_L \\ z_L \end{bmatrix} = |\vec{L}| \cdot \begin{bmatrix} \sin\alpha\cos\rho \\ \sin\alpha\sin\rho \\ \cos\alpha \end{bmatrix}. \tag{33}$$

Let $[x_L, y_L, z_L]^T$ be the coordinate of the laser footprint in the instantaneous laser scanning coordinate system, α be the zenith angle between the laser propagation direction and the vertical direction, ρ be the azimuth angle of the laser projection on the plane, and $\vec{L}$ be the laser propagation vector above the water surface as (34), shown at the bottom of the page. The positioning deviation of the laser scanning system in the IMU coordinate system is $[\,\Delta x_{\mathrm{LI}}\ \Delta y_{\mathrm{LI}}\ \Delta z_{\mathrm{LI}}\,]^T$. The position measurement of the devices and system calibration will help obtain the precise value of $[\,\Delta x_{\mathrm{LI}}\ \Delta y_{\mathrm{LI}}\ \Delta z_{\mathrm{LI}}\,]^T$. R_{LI} is the rotation matrix of the laser scanning coordinate system to the inertial platform reference coordinate system. R_{IG} is constructed based on three real-time observations, heading, pitch, and roll from the IMU. $[\,\Delta x_{\mathrm{GI}}\ \Delta y_{\mathrm{GI}}\ \Delta z_{\mathrm{GI}}\,]^T$ is the position deviation from the reference center of the inertial platform to the phase center of the global navigation satellite system (GNSS) antenna. In addition, R_{GW} is the rotation matrix for the conversion to the geodetic coordinate system. The rotation angle of R_{GW} needs to refer to the latitude and longitude in the geodetic coordinate system. $[\,\delta x\ \delta y\ \delta z\,]^T$ is the spatial position of the GNSS antenna phase center in the geodetic coordinate system. The angle and composition of the three rotation matrices R_{LI}, R_{IG}, and R_{GW} should be determined according to the installation direction of each system.

After the above calculation, the real-time point position detected by the system is converted into the geodetic coordinate system, and then, the calculation of the 3-D position of the laser foot point space is completed.

2) Below the Water Surface: When the ALB system detects near-shore water and land targets, the laser propagation process is often more complicated than that of a conventional LiDAR. This is mainly because the ranging laser has to pass successively through two nonuniform media of air and water during the ranging laser detection process. Therefore, the position of the laser spot needs to be corrected for the propagation direction and the propagation speed [46]. According to Snell's law, the propagation direction and speed of a laser have the following relationship:

$$\frac{\sin\alpha}{\sin\beta} = \frac{c}{v} = \frac{n_1}{n_2} = n_w \tag{35}$$

where α is the zenith angle, β is the corresponding refraction angle, v and c are the speed of light in water and air, respectively, and n_1 and n_2 are the indexes of the laser in water and air, respectively.

As shown in Fig. 7, according to echo signal processing, the propagation time of a laser in water can be obtained by calibrating the position of the echo on the water surface, and the propagation distance of a laser under the water surface is represented as follows:

$$|\vec{F}| = |P_s P_{\mathrm{uw}}| = \mathrm{vt}. \tag{36}$$

$$P_s = \begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = R_{\mathrm{GW}}\left[R_{\mathrm{IG}}\left[R_{\mathrm{LI}}\begin{bmatrix} x_L \\ y_L \\ z_L \end{bmatrix} + \begin{bmatrix} \Delta x_{\mathrm{LI}} \\ \Delta y_{\mathrm{LI}} \\ \Delta z_{\mathrm{LI}} \end{bmatrix}\right] + \begin{bmatrix} \Delta x_{\mathrm{IG}} \\ \Delta y_{\mathrm{IG}} \\ \Delta z_{\mathrm{IG}} \end{bmatrix}\right] + \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix} \tag{34}$$

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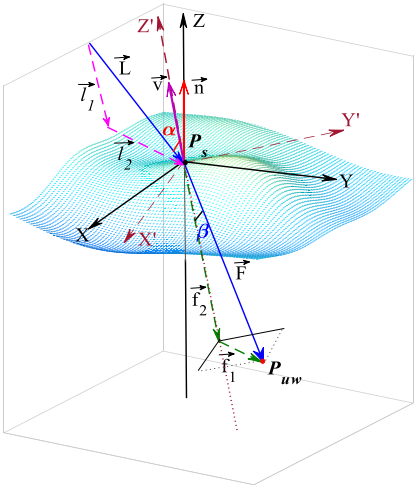

Fig. 7. Illustration of the refraction process of a laser under sea surface fluctuation.

To locate a target in the geographic plane, we also need to know the direction of underwater light propagation. Its vector form is given as follows:

$$\vec{F} = \left(\frac{\vec{L}}{|\vec{L}| \cdot n_w} - \vec{n} \cdot \sqrt{1 - \frac{1 - \cos^2 \alpha}{n_w^2}} \right) \cdot |\vec{F}| \quad (37)$$

$$P_{\text{uw}} = P_s + \vec{F}. \quad (38)$$

Their values can be calculated according to the method [47]. On this basis, the position of the underwater point cloud can be calculated according to formula (38).

III. Experiment

The data processing method proposed in this article aims mainly to solve the problem of target echo response detection in the received full-waveform signal by an ALB system. The necessity of target echo identification will be pronounced in some coastal areas with complex scattering interference in optical transmission media. Based on the method proposed in this article, we use the iGreena system developed by Shenzhen University [48] and the Aquarius system produced by Teledyne Optech [49] to take the full-waveform echo signals in the coastal areas for data processing experiments (Fig. 8). Table II shows the parameters of the system used in the experiment. By comparing and analyzing the effect before and after data processing, the feasibility of the echo algorithm strategy in airborne bathymetric LiDAR signal processing is analyzed. The data processing experiment focuses on the following two aspects: 1) comparison of the difference between a conventional echo signal decomposition, point cloud filtering, and amplification of the CRF methods and 2) verification of the effect of a CRF model on echo signal processing and its practical significance.

A. Study Area

To verify the general applicability of this method, we selected two coastal areas as study areas (Fig. 9). Wuzhizhou is an island in Hainan Province, China. The water depth is less than 20 m in the study area. The seafloor

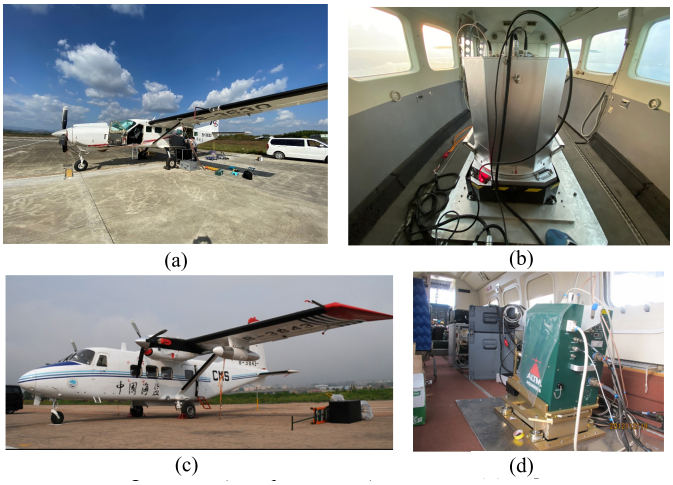

Fig. 8. Aircraft used in the experiment and ALB systems installation [47]. (a) Fixed-wing aircraft (Wuzhizhou). (b) iGreena. (c) Fixed-wing aircraft (Ganquan Island). (d) Aquarius.

TABLE II
Parameters of the Experimental Systems

System Parameters	iGreena	Aquarius
Wavelength	532 nm	532 nm
Pulse repetition rate	50~700 kHz	33~70 kHz
Scan path	Circular	Linear
Laser divergence angle	0.438mrad	1mrad
Acquisition frequency	1.25 GHz	1 GHz

surrounding Wuzhizhou is mainly composed of sand and organic debris sediments. In January 2020, we implemented a flight experiment over eastern Hainan Island based on the iGreena ALB system developed by the authors at Shenzhen University. Another study area is located north of Ganquan Island, which is an island in the South China Sea. Its sediment mainly consists of coral reefs, coral fragments, and debris. At the end of December 2012, the First Institute of Oceanography (FIO), Ministry of Natural Resources of China, carried out a measurement experiment around Ganquan Island by using the Aquarius system which is also a single-wavelength ALB system produced by Teledyne Optech [49]. The Secchi disk transparency depth in the two study areas is deeper than 8 m during the experiment. Therefore, the environmental conditions in the study areas are suitable for the use of ALB systems to carry out integrated land and water exploration. The bathymetry survey range can reach a depth of 2 Secchi in the two study areas. The HD-370 and HD-Max are single-beam echo sounders (SBESs) produced by Hi-Target Company, and their operating frequency is 200 kHz [50], [51]. We implemented a simultaneous bathymetric survey using the two SBESs in Wuzhizhou and Ganquan Island to assess the bathymetric accuracy of the different echo signal detection methods. The deviations between the main survey line and check lines in the single-beam bathymetric data were less than 0.3 m. Fig. 9 shows the distribution of the surveying points.

The study areas in this section contain a wide range of target types, which is conducive for the verification of the universal validity and applicability of the method proposed

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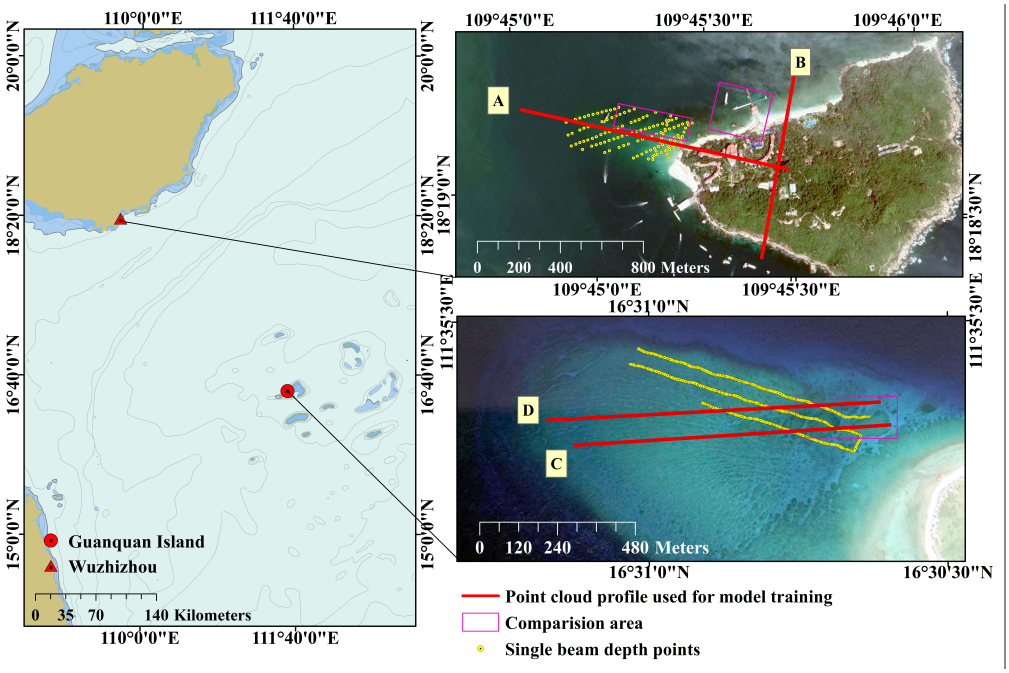

Fig. 9. Study area location and model training profile.

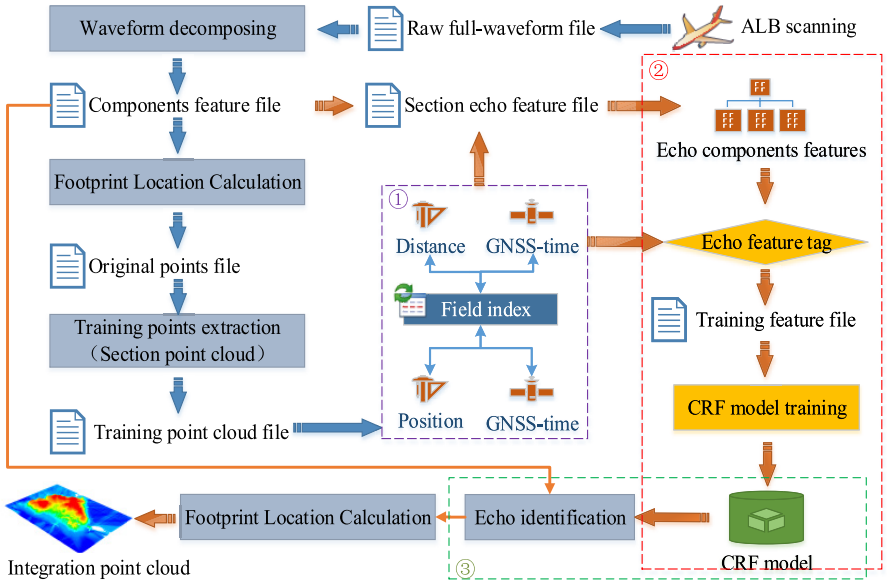

Fig. 10. Experimental data processing: ① Training data index; ② CRF model training; and ③ Target echo identification.

in this article. The point cloud data and the corresponding echo waveforms in the training profiles were extracted as the training dataset for constructing the CRF model.

B. Data Processing

In experimental data processing, the original footprints are generated using raw full-waveform data obtained using ALB scanning through waveform decomposition and location calculation. On this basis, a point cloud profile can be selected manually as the training data for the CRF model. The profile of a point cloud is convenient for observing the spatial structure characteristics and is beneficial for extracting target echo positions at different altitude levels. Moreover, the selected point cloud profile can run through the land and water region and contain most of the target types in the study area, improving the stability of the CRF model. In practical operation, man–machine interaction helps to realize the extraction of training data combined with other detection results. For example, a photographic image collected synchronously can provide some reference information for target identification. Fig. 10 shows the overall process of CRF model construction and data processing in this article.

A field index based on GNSS observation data is established between the training point cloud and its corresponding echo signal feature file. Field index ① makes it convenient to mark the target echo component from the component feature file after waveform decomposition. According to the CRF model, echo signal processing is divided into two steps: ② CRF model training and ③ target echo identification.

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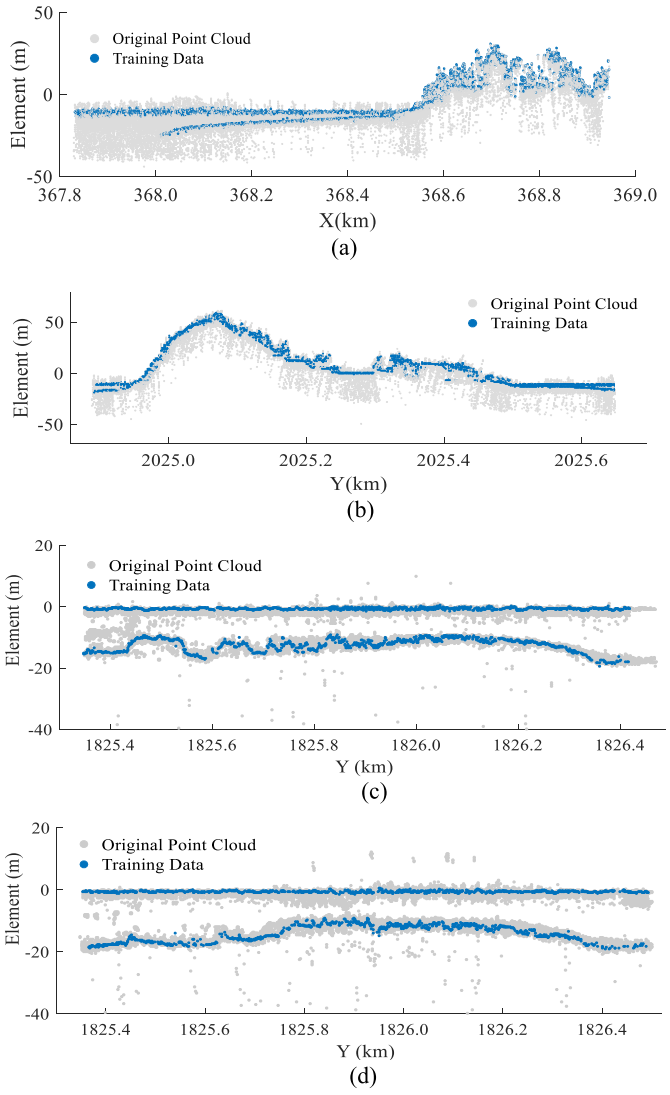

Fig. 11. Point cloud profiles for training. (a) Training Profile A (1094 beams). (b) Training Profile B (1294 beams). (c) Training Profile C (973 beams). (d) Training Profile D (925 beams).

1) Training Dataset: The establishment of the training dataset relies on extracting point cloud profiles containing multiple targets and their corresponding waveform data from the scanning data. The selection of training data should cover all-terrain features and objects as much as possible to improve the accuracy of model parameter estimation. For the study area in this article, the spatial positions of the profiles we selected are shown in Fig. 9. In Fig. 11, the original point cloud refers to the corresponding spatial positions of all Gaussian echo components after progressive Gaussian decomposition, which does not include signal component identification processing.

Fig. 11 shows four profiles selected for CRF model training in both experimental areas, in which the gray points represent the original point cloud. As observed from Fig. 11, if the echo components are not discriminated during the echo signal processing, the point cloud obtained after the spatial position calculation will contain a large number of nontarget points, increasing the uncertainty of the target surface detection result. The blue points are included in the training datasets that are extracted by using manual interaction. In the figure, the density of gray point clouds in the water body of Ganquan Island is significantly smaller than that of Wuzhizhou. This is because Ganquan Island is far away from the mainland and is less affected by water–land interaction and human activities so that the scattering in the water suffers less interference than that of Wuzhizhou.

2) CRF Model Accuracy Evaluation: Generally, there are obvious differences between the results of CRF processing using different feature functions. To analyze the effects of different models and evaluate the accuracy of the trained CRF model, two indicators are constructed here: the waveform components error rate (CER) and the laser beams error rate (LBER). The CER can reflect the accuracy of each waveform component

$$\mathrm{CER} = \frac{N_{c-\mathrm{sucess}}}{N_{\mathrm{comp}}} \times 100\%. \tag{39}$$

The LBER can be used to measure the change in the overall data in the detection area after waveform processing based on target echo identification

$$\mathrm{LBER} = \frac{N_{L-\mathrm{sucess}}}{N_{\mathrm{beams}}} \times 100\% \tag{40}$$

where N_{comp} represents the total number of echo components that need to be labeled in the training data, $N_{c-\mathrm{sucess}}$ is the number of echo components for which the labeling results are inconsistent with the training data, and N_{beams} is the total number of beams or the total number of waveforms in the training data and represents the waveform data whose labeled result is different from the position and number of echoes of the corresponding training data.

IV. Result and Discussion

A. Modeling Accuracy

We conducted CRF modeling based on the training dataset abstracted from the study areas, reapplied the target echo identification model to the profile data processing, and generated the corresponding point cloud. In this section, we compared the target echo selection based on the unary potential echo feature and multifeature sequential discrimination (MFSD) [9], [52] with the CRF model to verify the modeling effect. The MFSD is a method based on waveform decomposition and feature analysis. Various waveform features of echo components are regarded as mutually independent discriminant conditions, and all echo components obtained after waveform decomposition will be distinguished sequentially. To avoid redundancy of the echo component features that will increase the complexity of the calculation model and reduce the processing efficiency, we select unary potential features directly related to the shape of the signal waveform components, namely, signal width, amplitude, and the similarity with the transmitted pulse as the discriminant. The accuracy of the models within different conditions is analyzed by cross validation, as shown in Figs. 12–15 and Table III.

Figs. 12(a)–(c) to 15(a)–(c) show the processing results of unary potential conditions, that is, when the value of S_1, S_2, and S_3 is 1, the corresponding echo components will be considered as the target echo.

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TABLE III
TARGET ECHO IDENTIFICATION ERROR RATE WITH RESPECT TO DIFFERENT COMPONENT DISCRIMINANT STRATEGIES

Profile	A		B		C		D	
	CER(%)	LBER(%)	CER(%)	LBER(%)	CER(%)	LBER(%)	CER(%)	LBER(%)
	Unary potential identification							
Width $\{S_1 = 1\}$	34.67	99.49	26.51	97.15	26.17	83.25	22.81	82.17
Amplitude $\{S_2 = 1\}$	25.63	91.5	22.83	87.2	20.81	79.65	21.1	80.14
Similarity $\{S_3 = 1\}$	24.25	92.43	13.15	63.42	20.79	79.86	15.209	61.73
Multi-Feature Sequential Discrimination (MFSD)	22.55	91.06	20.39	87.34	18.95	72.46	15.187	61.43
	Conditional Random Fields							
$P\{S_1, S_2, T\}$	19.85	82.54	20.28	83.15	18.31	68.24	14.615	59.68
$P\{S_1, S_2, S_3, T\}$	15.64	69.24	9.12	51.8	16.52	61.05	14.59	59.57

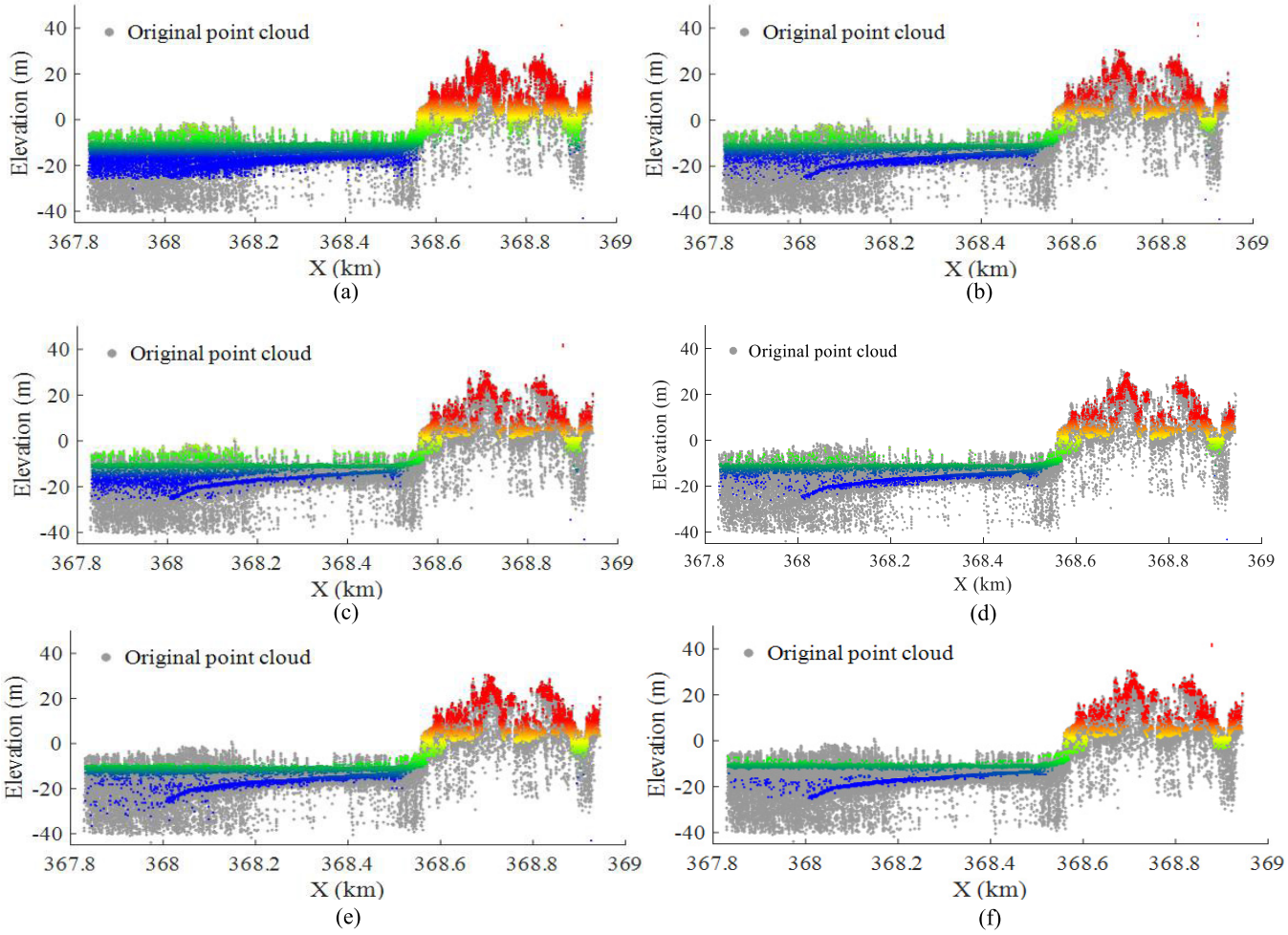

Fig. 12. Point cloud profiles from different identification methods (Profile A). (a) Width. (b) Amplitude. (c) Waveform similarity. (d) Multifeatures sequential discrimination. (e) CRF ($P\{S_1,S_2,T\}$). (f) CRF ($P\{S_1,S_2,S_3,T\}$).

In Table III, the CERs and LBERs show that the methods based on the unary potential features can improve the accuracy of target echo signal extraction. This verifies the importance of unary potential features in signal processing.

Among the unary potential features of each echo signal, the similarity feature associates the echo signal features with the laser transmitted pulses of the particular system. Therefore, the conditional detection error rate that considers the similarity feature is lower than that of other unary features. This phenomenon can also be found in the case of the CRF model processing based on the feature condition.

Figs. 12(d)–(f) to 15(d)–(f) show the results of multifeature combination identification. It is observed that the thickness of the surface point clouds, such as the water surface and the

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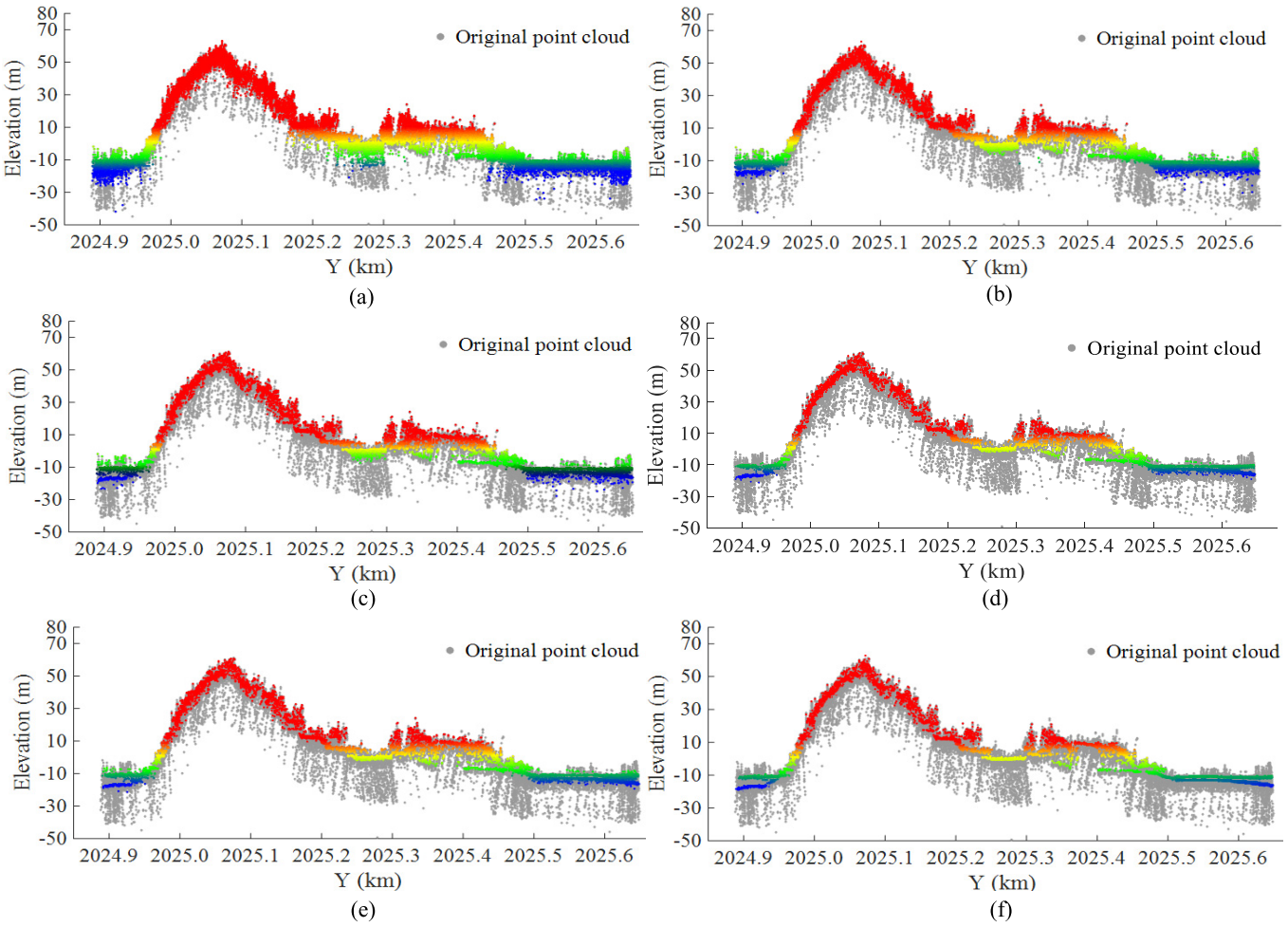

Fig. 13. Point cloud profiles from different identification methods (Profile B). (a) Width. (b) Amplitude. (c) Waveform similarity. (d) Multifeatures sequential discrimination. (e) CRF ($P\{S_1, S_2, T\}$) (f) CRF ($P\{S_1, S_2, S_3, T\}$).

seabed, decreases after the multifeature combination screening. The processing results presented in Figs. 12 and 13 indicate that the corresponding approach can reduce the uncertainty of the target surface identification. Moreover, the point cloud density in the water body is lower than that of the MFSD processing method. Therefore, the target echo identification with the CRF model training can avoid the negative effect of misjudging nontarget echo signals close to the target surface underwater.

In summary, through the quantitative comparison of the processing results of several methods in the experiment, it can be found that the discrimination error of the CRF model method is smaller than that of the single condition or the MFSD method in Table III. The profile point cloud generated after the echo signal distinction also shows that the CRF model trained by the actual area has a good processing effect.

In the experiment, we use C++ and MATLAB for joint programming to implement related algorithms and then record the time spent on the CRF model training and echo component identification in Table IV.

Although the time cost for each profile processing is relatively short, the time spent in the overall processing of large target areas cannot be ignored. In particular, Table IV does not include the training data processing part, which will obviously increase the processing time.

TABLE IV

TIME SPENT ON THE CRF MODEL TRAINING [② IN FIG. 9] AND ECHO COMPONENT IDENTIFICATION [③ IN FIG. 9]

Profiles	Number of waveforms	CRF model	Data processing stage (s)		
			②	③	②+③
A	1094	$P\{S_1, S_2, T\}$	0.08	0.84	0.92
		$P\{S_1, S_2, S_3, T\}$	0.09	0.88	0.97
B	1294	$P\{S_1, S_2, T\}$	0.08	0.82	0.9
		$P\{S_1, S_2, S_3, T\}$	0.08	0.85	0.93
C	973	$P\{S_1, S_2, T\}$	0.06	0.71	0.77
		$P\{S_1, S_2, S_3, T\}$	0.08	0.73	0.81
D	925	$P\{S_1, S_2, T\}$	0.06	0.69	0.75
		$P\{S_1, S_2, S_3, T\}$	0.07	0.74	0.81

The CRF model is introduced in this article to improve the accuracy of the target echo temporal position extraction and reduce the intervention of artificial operations in data processing. Therefore, the time cost complexity of computer programs is a critical factor to be considered. We believe that the optimization of the computer program and the realization of automatic extraction of training data are the key to method feasibility and are another important research direction in the future.

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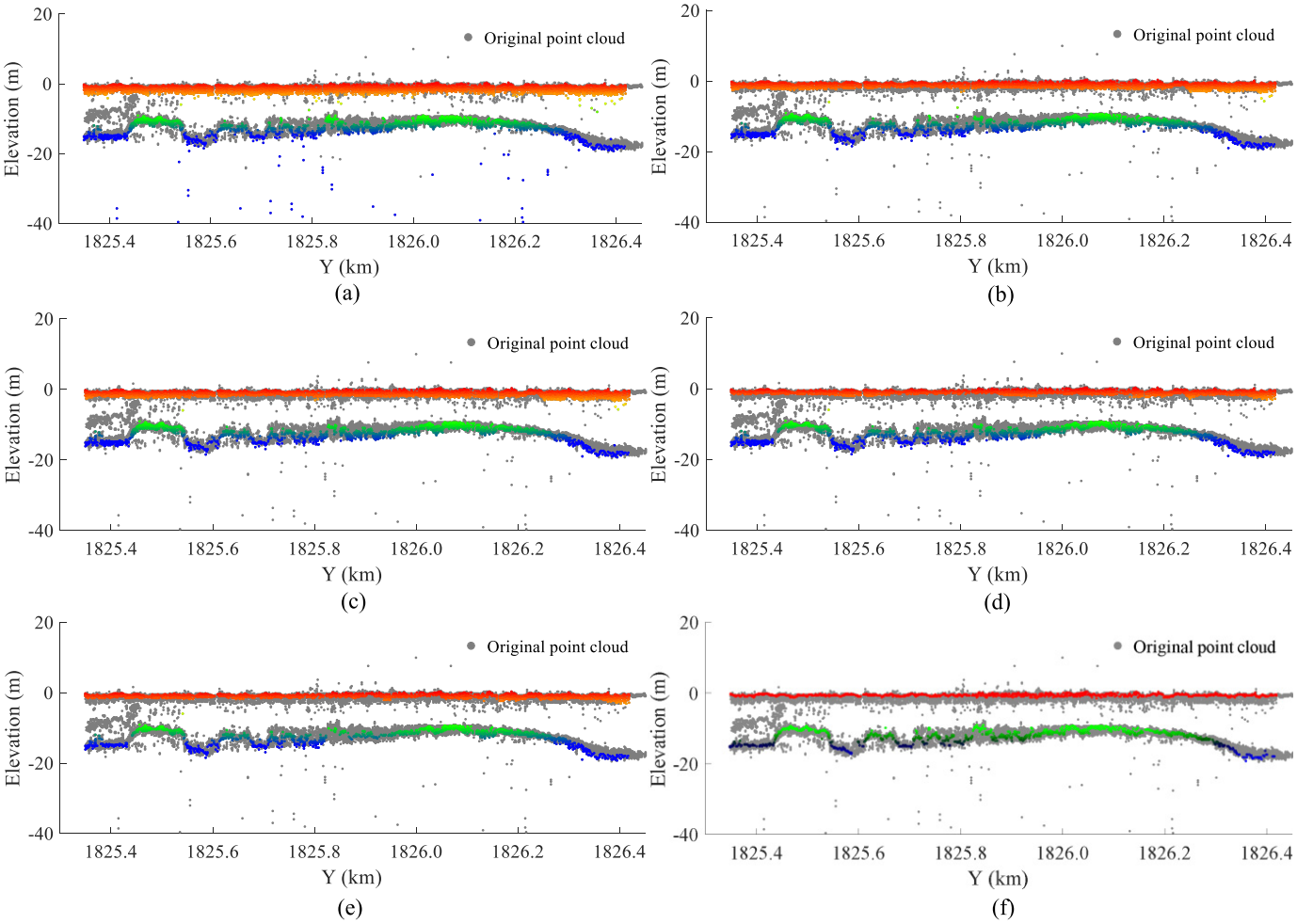

Fig. 14. Point cloud profiles from different identification methods (Profile C). (a) Width. (b) Amplitude. (c) Waveform similarity. (d) Multifeatures sequential discrimination. (e) CRF ($P\{S_1, S_2, T\}$). (f) CRF ($P\{S_1, S_2, S_3, T\}$).

B. Regional Application

In this section, we analyzed the CRF model effects on coastal and water areas. The point cloud processing results of the intensity threshold method were used for a comparison to study the processing effect of the echo component discrimination method after waveform decomposition in Figs. 16–18, which corresponds to the comparison area in Fig. 9. Then, we used the statistical outlier removal (SOR) method to filter the results of the three approaches [53]. K is the number of k-nearest points and n_k is the multiple parameter of the standard deviation. The processing effect is given as follows.

In Fig. 16, the point cloud obtained by directly using the intensity threshold method contains a large number of nontarget noise points. These points occur mainly due to the scattering interference in the propagation path. The topography in the target area has spatial continuity. After SOR filtering, the processing effect of point clouds with prominent spatial structure characteristics, such as buildings, seas, and beaches, will be pronounced. However, the filtering effect for the nontarget noise points in areas such as trees and shrubs with multielevation distribution in the vertical direction will always be insufficient. Furthermore, the interference of nontarget noise in Fig. 16(c) and (e) is significantly lower than that of the intensity threshold detection method. This result effectively verifies the positive role of the echo component discrimination method based on waveform decomposition in point cloud generation. Particularly for multielevation targets, the distinguishing effect between the canopy of plants and the ground is also visible.

It is observed from the comparison of Fig. 16(c) and (e) that the point cloud extracted based on the CRF model has fewer random noise points near the echo surface. The relevant point clouds show clearer structural characteristics, providing a better data basis for the subsequent point cloud spatial filtering processing.

In fact, the improvement of the data processing effect of the method proposed in this article is more pronounced in underwater terrain detection. Figs. 17 and 18 show the processing results on underwater terrain in the same scanning belt. Among them, the strong scattering in the water body interferes severely with the practical effect of peak detection [Figs. 17(a) and 18(a)]. Although the terrain changes of the bottom terrain can initially be observed after SOR processing, the whole area still contains a large density of random noise [Figs. 17(b) and 18(b)]. After processing with the CRF-based target signal screening model, the topography of the water bottom can be observed in Figs. 17(e) and 18(e). Since the continuous distribution of the bottom topography is already

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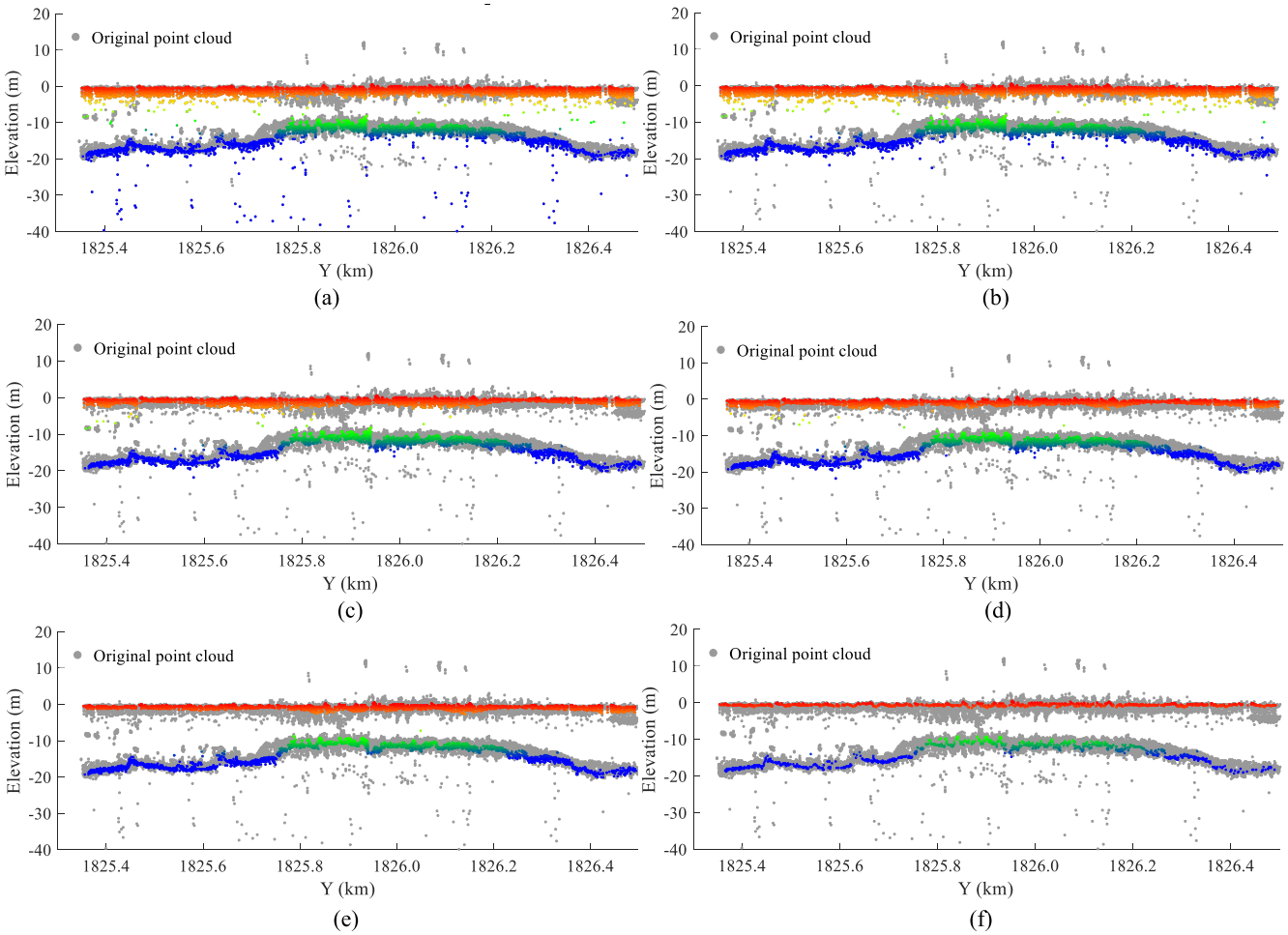

Fig. 15. Point cloud profiles from different identification methods (Profile D). (a) Width. (b) Amplitude. (c) Waveform similarity. (d) Multifeatures sequential discrimination. (e) CRF ($P\{S_1, S_2, T\}$). (f) CRF ($P\{S_1, S_2, S_3, T\}$).

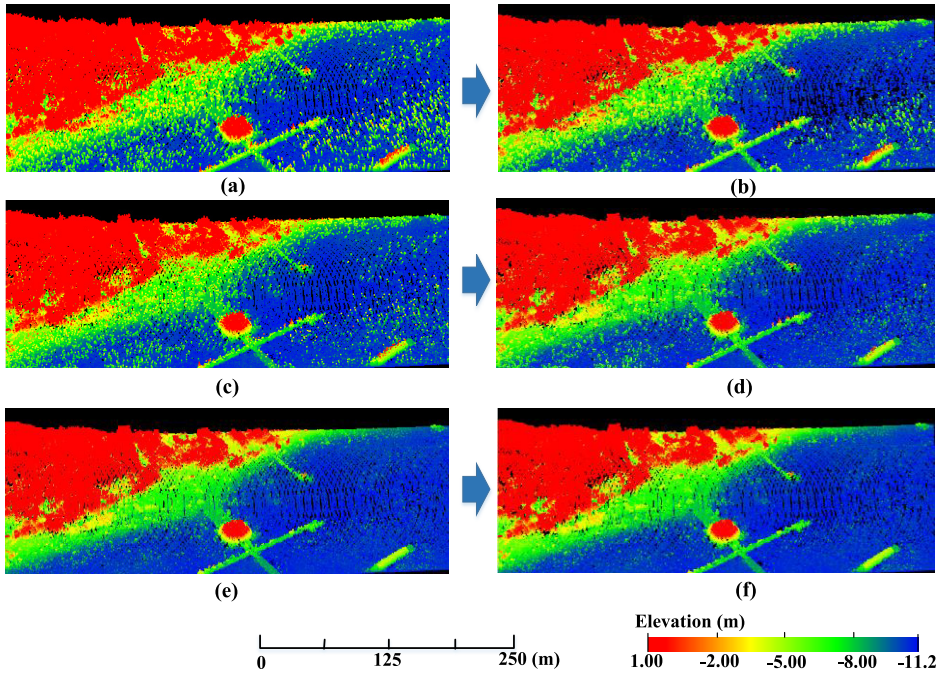

Fig. 16. Comparison of the processing effects in the coastal area (Wuzhizhou). (a) Echo detection based on signal intensity ($>3\sigma$). (b), (d), and (f) SOR filter ($K = 10$ and nk $= 3$). (c) Multifeatures sequential discrimination. (e) Echo detection with the CRF model.

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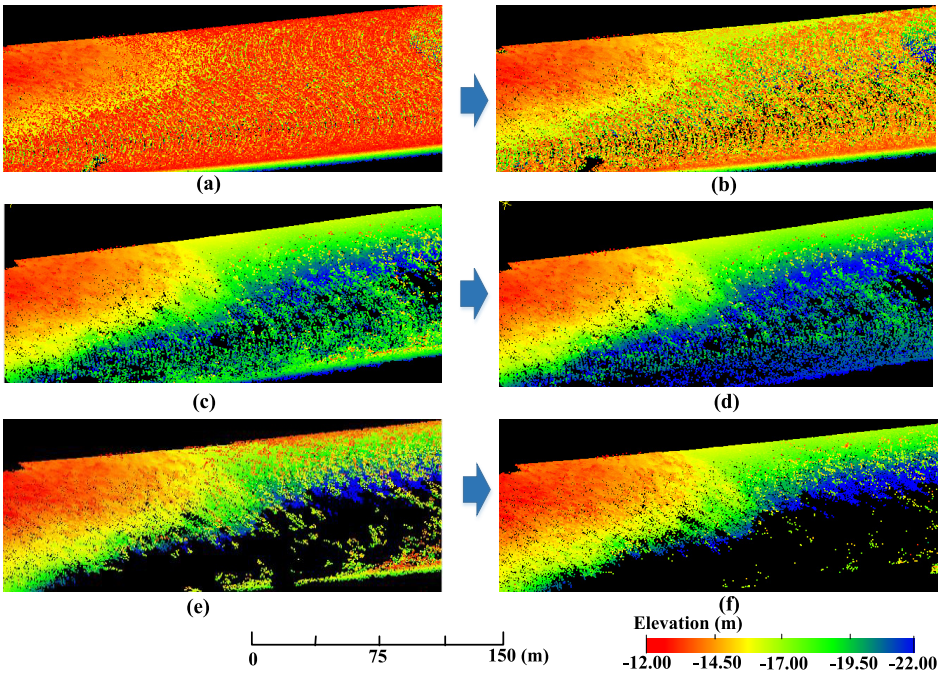

Fig. 17. Comparison of the processing effects in the water area (Wuzhizhou). (a) Echo detection based on signal intensity ($>3\sigma$). (b), (d), and (f) SOR filter ($K = 10$ and nk $= 3$). (c) Multifeatures sequential discrimination. (e) Echo detection with the CRF model.

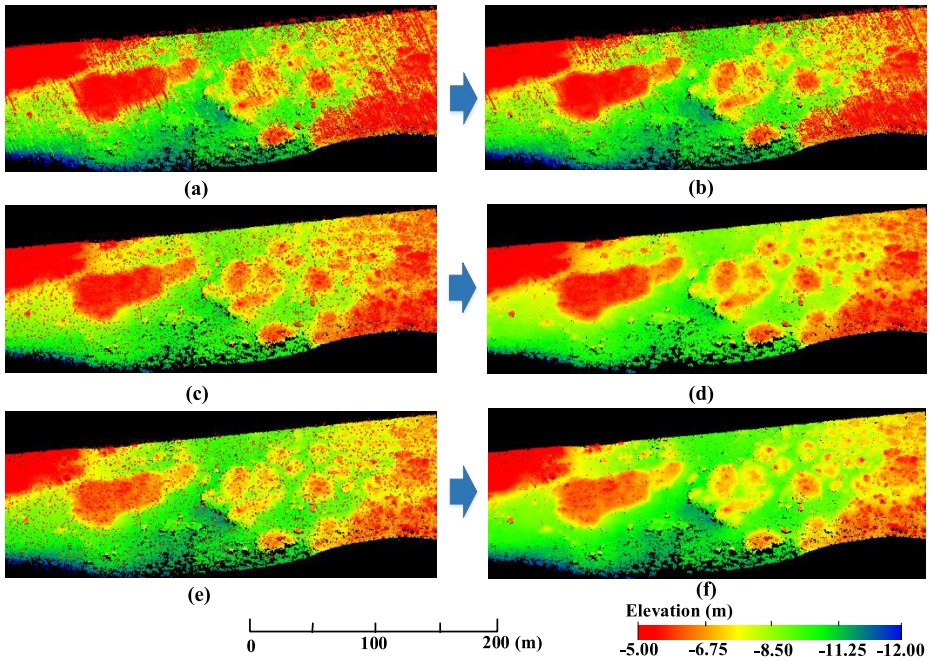

Fig. 18. Comparison of the processing effects in the water area (Ganquan Island). (a) Echo detection based on signal intensity ($>3\sigma$). (b), (d), and (f) SOR filter ($K = 10$ and nk $= 3$). (c) Multifeatures sequential discrimination. (e) Echo detection with the CRF model.

apparent, the SOR method has significantly reduced the non-target interference in the water body [Figs. 17(f) and 18(f)].

Fig. 19 shows a comparison of the probability density distribution of the point cloud elevation obtained by the system before and after using the method proposed in this article. Fig. 19(a) shows the elevation distribution of the water surface and coastal point clouds. Fig. 19(b) and (c) shows the elevation distribution of the underwater point cloud. Although the MFSD-based processing can effectively obtain the detection result of the target area to some extent, the curve of the elevation probability density of the detection result is relatively smooth. This is mainly because the point cloud contains considerable random noise and the target scattering section. The elevation distribution curve changes quickly, and there are multiple apparent peaks after adopting the echo component identification method proposed in this article. This is because the random noise is reduced in the results.

C. Elevation Detection Accuracy Underwater

The impact due to the different data processing methods will ultimately affect the precision of underwater terrain detection.

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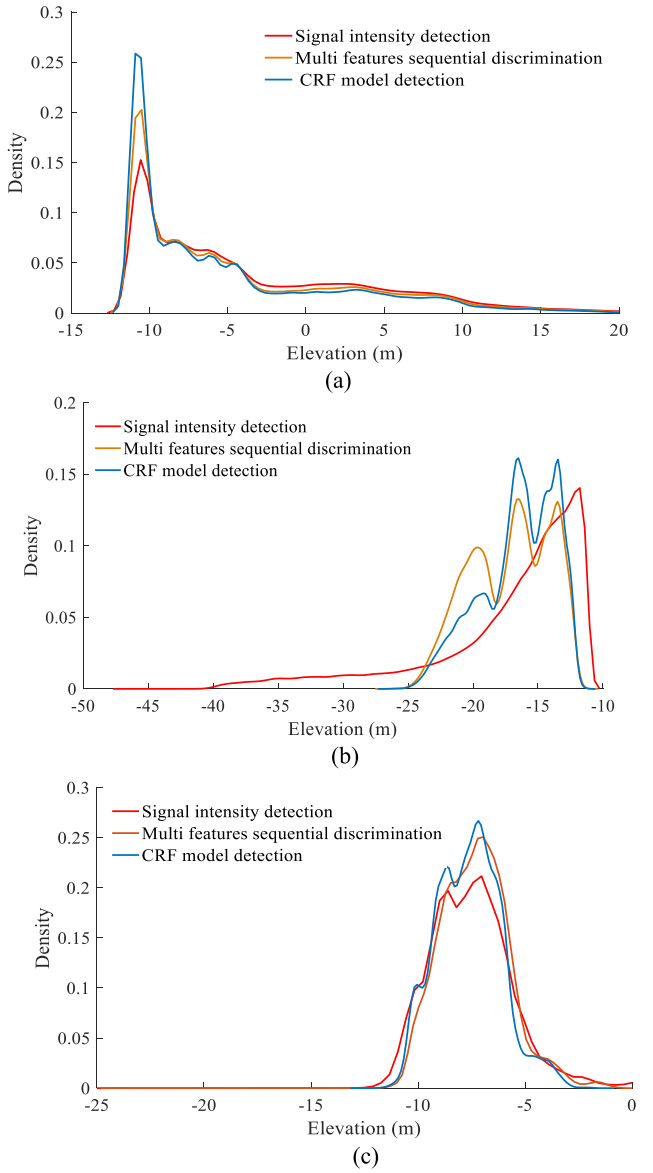

Fig. 19. Probability density distribution of the point cloud elevation by using different methods in the land or water area. (a) Coastal area (Wuzhizhou). (b) Water area (Wuzhizhou). (c) Water area (Ganquan Island).

Since the SBES is widely used marine exploration equipment with high bathymetric detection accuracy and stable performance, we regard the result of the elevation measurements as a reference. In this section, we assessed the accuracy of target echo component detection by comparing the depth deviation between the SBES and the different echo signal identification methods.

The distribution of SBES bathymetric surveying points in the study areas is shown in Fig. 9. We selected the ten nearest laser footprints from the ALB point cloud and used the inverse-distance weighting (IDW) algorithm to interpolate the depth values at the corresponding positions. Figs. 20 and 21 show the deviation statistics between the ALB underwater point cloud and the single-beam sounding points. The ALB underwater point clouds are generated by detecting the target echo components based on the MFSD and CRF methods.

TABLE V
ELEVATION DEVIATION STATISTICS BETWEEN THE ALB SYSTEM AND THE SBES SYSTEM

Area	Wuzhizhou		Ganquan Island	
Statistics	MFSD	CRF	MFSD	CRF
MAE (m)	0.151	0.142	0.183	0.134
RMSE (m)	0.201	0.169	0.210	0.163

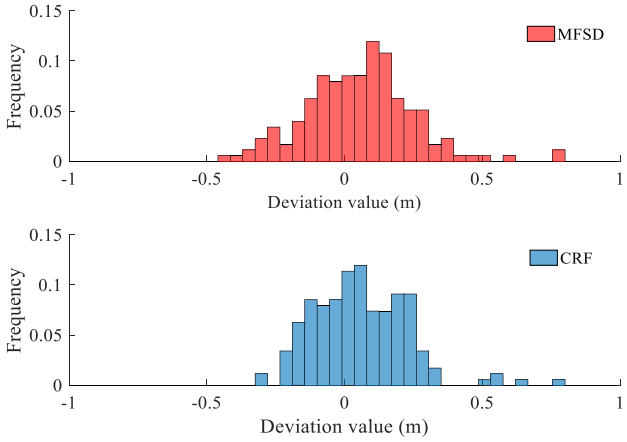

Fig. 20. Frequency of the depth deviation between the ALB system and the SBES in Wuzhizhou.

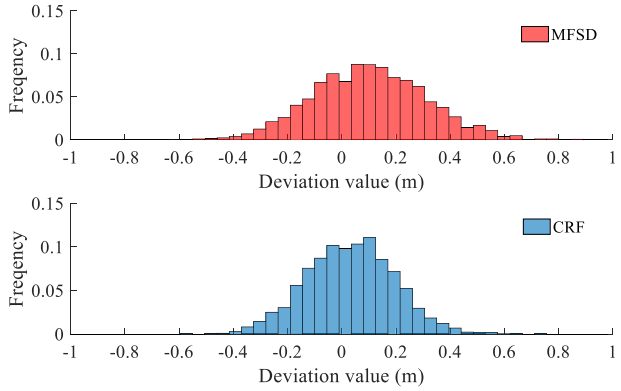

Fig. 21. Frequency of the depth deviation between the ALB system and the SBES in Ganquan Island.

The histograms of the deviation values from the different waveform processing methods exhibit a randomness characteristic, which is approximate Gaussian distribution. The deviation value generated from the method based on the CRF model is more concentrated with the small deviation, which means that the number of the smaller deviations is increasing. Table V lists the specific statistical values of the elevation deviation.

Introducing the CRF model into the full-waveform processing can reduce the uncertainty of the target surface caused by the detection error, and the elevation deviation of CRF-based processing is smaller than that of the MFSD. In Table V, the mean absolute error (MAE) of the water depth deviation decreased by 0.009 m in Wuzhizhou and 0.049 m in Ganquan Island. Moreover, the root-mean-squared error (RMSE) of the water depth deviation decreased by 0.032 m in Wuzhizhou and

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0.047 m in Ganquan Island. This indicates that the elevation detection accuracy of the MFSD-based target echo component detection is lower than that of the CRF-based method. In summary, the experiments above verify the effectiveness of the proposed methods.

Based on the several experiments described above, we found that the detection accuracy of the target echo signal based on the CRF model is significantly improved. The introduction of the CRF model into the full-waveform processing can reduce the uncertainty of the target surface caused by the detection error. The uncertainty manifests itself as an increase in the thickness of the surface point cloud. The above experiment verified the effectiveness of the proposed methods.

V. Conclusion

In the process of laser propagation, the strong scattering from an inhomogeneous medium and the complex structure of some backscattered target surfaces in the elevation direction will give rise to severe interference of the nontarget echo signal, making it difficult to extract the target position. Therefore, to avoid the poor effect of spatial point cloud filtering in the presence of strong scattering during laser propagation or when the target spatial structure is not clear, we have attempted to design a method of echo signal processing for target echo signal extraction. In this method, we use waveform decomposition to decompose the full-waveform signal into several components. On this basis, echo component features are extracted to construct the CRF model, thereby suppressing nontarget interference in point cloud generation and improving the target detection accuracy. According to the comparative analysis of the experimental results, we obtain the following conclusions.

1) The interference of the noise signal and the superposition of echo components affects the accuracy of target echo extraction in full-waveform processing. According to waveform decomposition, the extraction issue for the target echo component can transform into a conditional identification of the echo component's response features and coupling between the adjacent signal components. This provides an approach for using the nonindependent echo component features to extract the target echo accurately.

2) In the case of complex backscattered surfaces and different propagation media, waveform signal processing based on the signal CRF model compensates for the shortcomings of the general method for the determined target position that relies on the topological characteristics of the point cloud spatial distribution. It does not require additional information for the point cloud and gives full play to the advantages of the full-waveform ALB system.

3) The combined method of the full-waveform CRF model and point cloud spatial filtering can effectively improve the detection effect of the ALB system in coastal areas. Particularly for shallow coastal water areas that are significantly affected by water scattering, the accuracy improvement of the detection effect for the ALB system will be more pronounced.

In addition, the method proposed in this article combines target detection with the intelligent processing of the echo signal. To effectively improve the detection effect of an ALB system, it is important to reduce manual intervention in the study of online signal processing for the ALB system. Indeed, this research has prospects for many further applications, such as the automatic selection of training data and the correlation between full-waveform signals and adjective spatial locations to improve the target detection accuracy. We will further carry out more quantitative tests and study the application of this method in the online processing of system-received signals.

Author Contributions

Kai Guo, Qingquan Li, Qingzhou Mao, Chisheng Wang, and Anlei Wu conceived and designed the experiments. Kai Guo, Jiasong Zhu, Yanxiong Liu, Yongzhong Ouyang, and Yikai Feng contributed equally to analyzing the data. Kai Guo performed the experiments and wrote this article.

Conflicts of Interest

The authors declare no conflict of interest.

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References

[1] A. Schmidt, F. Rottensteiner, and U. Sörgel, "Water-land-classification in coastal areas with full waveform LiDAR data," *Photogrammetrie-Fernerkundung-Geoinf.*, vol. 2013, no. 2, pp. 71–81, May 2013.

[2] G. C. Guenther, "Airborne laser hydrography: System design and performance factors," NOS, Rockville, MD, USA, NOAA Prof. Paper Ser., Mar. 1985. [Online]. Available: https://repository.library.noaa.gov/view/noaa/2811

[3] P. Westfeld, H.-G. Maas, K. Richter, and R. Weiß, "Analysis and correction of ocean wave pattern induced systematic coordinate errors in airborne LiDAR bathymetry," *ISPRS J. Photogramm. Remote Sens.*, vol. 128, pp. 314–325, Jun. 2017.

[4] H. M. Tulldahl and K. O. Steinvall, "Simulation of sea surface wave influence on small target detection with airborne laser depth sounding," *Appl. Opt.*, vol. 43, no. 12, pp. 2462–2483, 2004.

[5] F. Ackermann, "Airborne laser scanning: Present status and future expectations," *ISPRS J. Photogramm. Remote Sens.*, vol. 54, no. 2, pp. 64–67, Jul. 1999.

[6] C. Michoud et al., "Landslide detection and monitoring capability of boat-based mobile laser scanning along Dieppe coastal cliffs, Normandy," *Landslides*, vol. 12, no. 2, pp. 403–418, Apr. 2015.

[7] E. J. Huising and L. M. Gomes Pereira, "Errors and accuracy estimates of laser data acquired by various laser scanning systems for topographic applications," *ISPRS J. Photogramm. Remote Sens.*, vol. 53, no. 5, pp. 245–261, Oct. 1998.

[8] H. Abdallah, J.-S. Bailly, N. N. Baghdadi, N. Saint-Geours, and F. Fabre, "Potential of space-borne LiDAR sensors for global bathymetry in coastal and inland waters," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 1, pp. 202–216, Feb. 2013.

[9] K. Guo, W. Xu, Y. Liu, X. He, and Z. Tian, "Gaussian half-wavelength progressive decomposition method for waveform processing of airborne laser bathymetry," *Remote Sens.*, vol. 10, no. 2, p. 35, Dec. 2017.

[10] T. Allouis, J. S. Bailly, Y. Pastol, and C. Le Roux, "Comparison of LiDAR waveform processing methods for very shallow water bathymetry using Raman, near-infrared and green signals," *Earth Surf. Proc. Land.*, vol. 35, no. 6, pp. 640–650, May 2010.

[11] C. Wang, Q. Li, Y. Liu, G. Wu, P. Liu, and X. Ding, "A comparison of waveform processing algorithms for single-wavelength LiDAR bathymetry," *ISPRS J. Photogramm. Remote Sens.*, vol. 101, pp. 22–35, Mar. 2015.

[12] K. Y. Shrestha, W. E. Carter, K. C. Slatton, and T. K. Cossio, "Shallow bathymetric mapping via multistop single photoelectron sensitivity laser ranging," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4771–4790, Nov. 2012.

[13] J. Zhao, X. Zhao, H. Zhang, and F. Zhou, "Shallow water measurements using a single green laser corrected by building a near water surface penetration model," *Remote Sens.*, vol. 9, no. 5, p. 426, Apr. 2017.

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[14] C. Mulsow, "A flexible multi-media bundle approach," in *Proc. Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, Newcastle upon Tyne, U.K., 2010. [Online]. Available: https://www.isprs.org/PROCEEDINGS/XXXVIII/part5/papers/116.pdf
[15] J. H. Churnside and P. L. Donaghay, "Thin scattering layers observed by airborne lidar," *ICES J. Mar. Sci.*, vol. 66, no. 4, pp. 778–789, May 2009.
[16] M. A. Hofton, J. B. Minster, and J. B. Blair, "Decomposition of laser altimeter waveforms," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 4, pp. 1989–1996, Jul. 2000.
[17] R. Schwarz, N. Pfeifer, M. Pfennigbauer, and A. Ullrich, "Exponential decomposition with implicit deconvolution of lidar backscatter from the water column," *PFG–J. Photogramm., Remote Sens. Geoinf. Sci.*, vol. 85, no. 3, pp. 159–167, Aug. 2017.
[18] R. Schwarz, G. Mandlburger, M. Pfennigbauer, and N. Pfeifer, "Design and evaluation of a full-wave surface and bottom-detection algorithm for LiDAR bathymetry of very shallow waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 1–10, Apr. 2019.
[19] A. Chauve et al., "Processing full-waveform lidar data: Modelling raw signals," in *Proc. Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, 2007, pp. 1–6.
[20] A. R. W. Wagner and T. Melzer, "Waveform analysis techniques in airborne laser scanning," in *Proc. ISPRS Workshop Laser Scanning*, Espoo, Finland, 2007, pp. 413–418.
[21] K. Madsen, H. B. Nielsen, and O. Tingleff, *Methods for Non-Linear Least Squares Problems*, 2nd ed. Denmark, U.K.: Technical Univ. of Denmark, 2004. [Online]. Available: https://findit.dtu.dk/en/catalog/537f0d6e7401dbcc1200d44d
[22] G. Q. Zhou et al., "Comparison analysis of five waveform decomposition algorithms for the airborne LiDAR echo signal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7869–7880, 2021.
[23] K. Saylam, R. A. Brown, and J. R. Hupp, "Assessment of depth and turbidity with airborne lidar bathymetry and multiband satellite imagery in shallow water bodies of the Alaskan north slope," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 58, pp. 191–200, Jun. 2017.
[24] M. Liu et al., "Adaptive depth extraction algorithm for ocean lidar," *Chin. J. Lasers-Zhongguo Jiguang*, vol. 45, no. 10, Oct. 2018.
[25] C. Qi et al., "Fitting algorithm for airborne laser bathymetric waveforms based on layered heterogeneous model," *Infr. Laser Eng.*, vol. 48, no. 2, 2019, Art. no. 0206004.
[26] Y. Song, H. Li, and G. Zhai, "Comparative analysis of airborne laser bathymetric waveforms denoising algorithms," *Acta Geodaetica et Cartographica Sinica*, vol. 50, no. 2, pp. 270–278, 2021.
[27] X. Zhao, H. Xia, J. Zhao, and F. Zhou, "Adaptive wavelet threshold denoising for bathymetric laser full-waveforms with weak bottom returns," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[28] A. Collin, B. Long, and P. Archambault, "Merging land-marine realms: Spatial patterns of seamless coastal habitats using a multispectral LiDAR," *Remote Sens. Environ.*, vol. 123, pp. 390–399, Aug. 2012.
[29] N. Wilson et al., "Mapping seafloor relative reflectance and assessing coral reef morphology with EAARL-B topobathymetric LiDAR waveforms," *Estuaries Coasts*, vol. 45, pp. 923–937, Nov. 2019.
[30] D. Su, F. Yang, Y. Ma, K. Zhang, J. Huang, and M. Wang, "Classification of coral reefs in the South China Sea by combining airborne LiDAR bathymetry bottom waveforms and bathymetric features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 815–828, Feb. 2019.
[31] M. O. Gonsalves, "A comprehensive uncertainty analysis and method of geometric calibration for a circular scanning airborne lidar," Ph.D. dissertation, Univ. Southern Mississippi, Ann Arbor, MI, USA, 2010. [Online]. Available: https://www.proquest.com/docview/835075165?pq-origsite=gscholar&fromopenview=true
[32] A. Yang et al., "Filtering of airborne LiDAR bathymetry based on bidirectional cloth simulation," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 49–61, May 2020.
[33] Y. Chen, Z. Zhu, Y. Le, Z. Qiu, C. Gang, and L. Wang, "Refraction correction and coordinate displacement compensation in nearshore bathymetry using ICESat-2 LiDAR data and remote-sensing images," *Opt. Exp.*, vol. 29, no. 2, pp. 2411–2430, 2021.
[34] Z. Pan, C. Glennie, P. Hartzell, J. C. Fernandez-Diaz, C. J. Legleiter, and B. T. Overstreet, "Performance assessment of high resolution airborne full waveform LiDAR for shallow river bathymetry," *Remote Sens.*, vol. 7, no. 5, pp. 5133–5159, May 2015.
[35] Y. Ioannou, B. Taati, R. Harrap, and M. Greenspan, "Difference of normals as a multi-scale operator in unorganized point clouds," in *Proc. 2nd Int. Conf. 3D Imag., Model., Process., Vis. Transmiss.*, Oct. 2012, pp. 501–508.
[36] L. Rummelhard, A. Paigwar, A. Negre, and C. Laugier, "Ground estimation and point cloud segmentation using SpatioTemporal conditional random field," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2017, pp. 1105–1110.
[37] A. B. Jahromi, M. J. V. Zoej, A. Mohammadzadeh, and S. Sadeghian, "A novel filtering algorithm for bare-Earth extraction from airborne laser scanning data using an artificial neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 4, no. 4, pp. 836–843, Dec. 2011.
[38] G. Sithole and G. Vosselman, "Experimental comparison of filter algorithms for bare-Earth extraction from airborne laser scanning point clouds," *ISPRS J. Photogram. Remote Sens.*, vol. 59, nos. 1–2, pp. 85–101, Aug. 2004.
[39] J. H. Churnside, J. M. Sullivan, and M. S. Twardowski, "LiDAR extinction-to-backscatter ratio of the ocean," *Opt. Exp.*, vol. 22, no. 15, pp. 18698–18706, 2014.
[40] B. Cao et al., "An active-passive fusion strategy and accuracy evaluation for shallow water bathymetry based on ICESat-2 ATLAS laser point cloud and satellite remote sensing imagery," *Int. J. Remote Sens.*, vol. 42, no. 8, pp. 2783–2806, Apr. 2021.
[41] Y. Xiusong, and H. Motao, "Research on principle and data processing methods of airborne laser bathymetric technique," PLA Inf. Eng. Univ., Zhengzhou, China, 2010.
[42] M. Drton and M. H. Maathuis, "Structure learning in graphical modeling," *Annu. Rev. Statist. Appl.*, vol. 4, no. 1, pp. 365–393, 2017. [Online]. Available: https://www.annualreviews.org/doi/abs/10.1146/annurev-statistics-060116-053803
[43] P. Launeau et al., "Full-waveform LIDAR fast analysis of a moderately turbid bay in Western France," *Remote Sens.*, vol. 11, no. 2, p. 117, Jan. 2019.
[44] L. Abady, J.-S. Bailly, N. Baghdadi, Y. Pastol, and H. Abdallah, "Assessment of quadrilateral fitting of the water column contribution in LiDAR waveforms on bathymetry estimates," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 4, pp. 813–817, Apr. 2014.
[45] D. B. Nagle and C. W. Wright, "Algorithms used in the airborne lidar processing system (ALPS)," U.S. Geolog. Surv., Reston, VA, USA, Tech. Rep. 2016–1046, 2016. [Online]. Available: https://pubs.usgs.gov/of/2016/1046/ofr20161046.pdf
[46] E. Shen, Y. Zhang, and K. Li, "Positioning model and simulation of conical scanning airborne laser bathymetry system," *Chin. J. Lasers*, vol. 43, no. 2, 2016, Art. no. 0214001.
[47] K. Guo et al., "Errors of airborne bathymetry LiDAR detection caused by ocean waves and dimension-based laser incidence correction," *Remote Sens.*, vol. 13, no. 9, p. 1750, 2021.
[48] K. Guo et al., "Development of a single-wavelength airborne bathymetric LiDAR: System design and data processing," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 62–84, Mar. 2022.
[49] C. J. Legleiter, B. T. Overstreet, C. L. Glennie, Z. Pan, J. C. Fernandez-Diaz, and A. Singhania, "Evaluating the capabilities of the CASI hyperspectral imaging system and aquarius bathymetric LiDAR for measuring channel morphology in two distinct river environments," *Earth Surf. Processes Landforms*, vol. 41, no. 3, pp. 344–363, Mar. 2016.
[50] Hi-Target. *Operating Instructions for HD-310, 360, 370, 380, 390 Sounder V1.1*. Accessed: Oct. 25, 2022. [Online]. Available: https://www.zhdgps.com/search/&type_id=5&keywords=370#
[51] Hi-Target. *Equipment Parameters of Single Beam Bathymeter HD-MAX*. Accessed: Oct. 25, 2022. [Online]. Available: https://www.zhdgps.com/detail/sounder-HD_MAX
[52] Y.-C. Lin, J. P. Mills, and S. Smith-Voysey, "Rigorous pulse detection from full-waveform airborne laser scanning data," *Int. J. Remote Sens.*, vol. 31, no. 5, pp. 1303–1324, Mar. 2010.
[53] C. Ma, W. Meng, B. Wu, S. Xu, and X. Zhang, "Efficient joint gradient based attack against SOR defense for 3D point cloud classification," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1819–1827.

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